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Vision and research directions of 6G technologies and applications

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ABSTRACT

Fifth-generation (5G) mobile communication technology is now widely available in several countries, with millions of 5G customers. Therefore, it's time for academia and business to focus on the next generation. This paper will overview the sixth-generation (6G) mobile network, including motivations, use case scenarios, requirements, supported research projects, and technologies. We discuss the Beyond 5G (B5G) evolution and advanced 5G features to predict the critical 6G requirements and highlight the 6G capabilities. We also introduce the 6G scenarios, requirements, and technological components compared to 5G.

Moreover, the current status of 6G research is discussed, and a rough roadmap for specification and regulation is explored. Then we describe a few prospective applications, their benefits, concepts, and research directions. We explore the business direction for 6G by introducing the most recently 6G projects in the vertical markets. We also propose a network architectural vision and the evolution of hardware-software designs to satisfy the higher requirements of 6G applications. This paper also presents a comprehensive survey of existing 6G trends, technologies, applications, industrial markets, and network structures for the most promising 6G applications.

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Contents

1. Introduction	2420
1.1. 6G drivers and mobile traffic growing	2421
1.2. Paper contributions	2422
2. Evolution of earlier mobile networks	2423
2.1. Evolution from 1G to 4G	2423
2.2. Evolution of 5G	2423
2.2.1. 5G timeline	2423
2.2.2. 5G features, deployment modes, and plans	2424
2.3. Evolution of 6G	2425
2.4. New scenarios of 6G beyond 2030	2425
3. Existing 6G surveys	2426
4. 6G requirements and capabilities	2428
4.1. Network flexibility	2428

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4.2.	End-to-end communication	2428
4.3.	High coverage	2428
4.4.	Diversity of embedded devices	2429
4.5.	Cognitive networks	2429
4.6.	Network computing	2429
4.7.	Trusted systems	2429
5.	6G network architectural vision and hardware -Software evolution	2429
5.1.	6G architectural vision	2429
5.2.	6G future hardware and software perspective	2430
5.2.1.	HW-SW divide	2430
5.2.2.	HW-SW symmetry	2430
5.2.3.	HW-SW separation	2430
5.2.4.	MICRO-NANO technologies in 6G	2430
6.	B5G and 6G vertical industries projects	2431
6.1.	B5G projects	2431
6.1.1.	MARSAL project	2432
6.1.2.	TeraFlow project	2432
6.1.3.	DAEMON project	2432
6.1.4.	REINDEER project	2432
6.2.	6G projects	2433
6.2.1.	Hexa-X project	2433
6.2.2.	DEDICAT6 project	2433
6.2.3.	RISE 6G project	2433
6.2.4.	Meta project	2434
6.2.5.	6G BRAINS project	2434
7.	The 6G applications requirements and challenges	2434
7.1.	E-health	2434
7.2.	Drones (unmanned vehicles)	2435
7.3.	Factory of the future	2437
7.4.	Localization	2437
7.5.	Connected autonomous vehicles (CAV)	2438
8.	Conclusion	2439
	Declaration of Competing Interest	2440
	Acknowledgement	2440
	References	2440

1. Introduction

The worldwide implementation of fifth-generation (5G) radio networks was expected to be completed by 2020, with additional high capacity connections, extreme trustworthiness, and assured low latency. However, 5G will suffer from a lack of satisfying all the future applications' demands beyond 2030. In addition, to increase coverage and lower power consumption, the sixth generation (6G) mobile network is expected to provide a more extensive spectrum, cost-effectiveness, and a high level of security. Several technological advances, including waveform design, multiple access, channel coding methods, different antenna technologies, network slicing, and cloud edge computing, are used to address these requirements in a 6G network. 6G impacts four significant future developments keys (You et al., 2021; De Alwis et al., 2021). These keys include global coverage, various spectra, various new applications and services, and strong security. The first key provides a centralized air, ground, space, and sea communication network by merging terrestrial and non-terrestrial networks (Yazar et al., 2020; Ray et al., 2021), as shown in Fig. 1.

In contrast, the second key introduces additional radio bands, including sub-6 GHz, millimeter-wave (mmWave), optical communications, and terahertz (THz). These additional bands will improve the capability and efficiency of connectivity.

The third key represents the complete applications and services using artificial intelligence (AI) and big data technologies. Due to the vast datasets produced by diverse systems with different wireless access technologies, comprehensive bandwidth utilization and antenna designs are required for 6G technical specifications (Elmeadawy and Shubair, 2019; Bariah et al., 2020; Yang et al.,

2020; Sheth et al., 2020). The fourth key includes robust security solutions that provide upgrading and increase network confidentiality (Huang et al., 2019). New security solutions must be introduced to support the physical and network layer. A comprehensive presentation of the 6G requirements and applications and those of other forthcoming projects is provided in this paper. 6G networks have several challenges, with processing data, detection systems, network monitoring, and data security among the most urgent issues.

The Internet of Everything (IoE) introduces critical security challenges due to the diverse range of functionality and challenges deploying distributed artificial intelligence. Given the high degree of mobility experienced by these newly-connected devices, they often switch between interconnected networks and seek services from other networks, resulting in security difficulties and data privacy issues. It is expected that the end-to-end latency in 6G would be reduced to one millisecond or less than to reach a few microseconds for the Enhanced Ultra-Reliable and Low Latency Communication (EURLLC) services.

Furthermore, a ten-fold improvement in network energy efficiency over 5G and a hundred-fold increase over 4G will be required for 6G (Pereira and Sousa, 2004). It is expected to provide highly low-power transmissions for devices with limited resources. Advanced and proactive management systems will enable rapid movement at speeds of up to 1000 km per hour for high mobility. The delayed impact of security procedures will be examined to ensure that the quality of service for EURLLC is maintained. The same is valid for demanding customers who seek extremely effective security mechanisms that guarantee services and resources availability. The IoE presents challenges in implementing

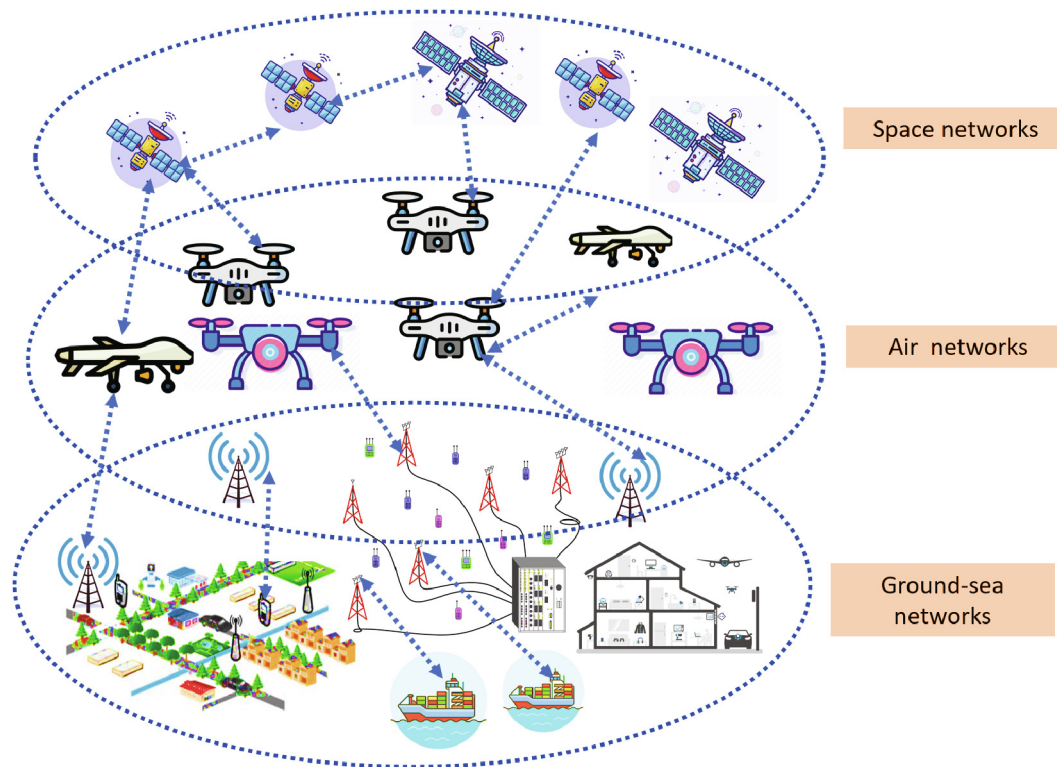


Fig. 1. The integrated ground-sea-air-space 6G expected network.

and managing new distributed intelligent AI and machine learning security mechanisms (Rupprecht et al., 2018; Hakeem et al., 2020).

1.1. 6G drivers and mobile traffic growing

A new generation of mobile communications has been launched virtually every decade. Advanced Mobile Phone System (AMPS) introduced the first generation (1G) analog cellular networks in the US and Nordic Mobile Telephone (NMT) in Europe circa 1980. Around 1990, second-generation (2G) digital cellular networks supplanted a 1G analog system. The Global System Mobile Communications (GSM) is a huge commercial success to allow over a billion people using mobile phone, text, and data services (Pelkmans, 2001). Third-generation (3G) systems (Dasilva et al., 1996) based on Code-Division Multiple Access (CDMA) were designed and initially implemented in 2001 to provide high-speed data access at several megabits per second. The world's first commercial Long-Term Evolution (LTE) networks launched in December 2009 in Stockholm and Oslo, delivering the world's first 4G mobile broadband service (Arshad et al., 2010). The 4G system encourages the growth of smartphones, spawning a trillion-dollar-a-year mobile Internet sector.

Finally, we entered the 5G era in April 2019. 5G extends mobile communication services beyond humans to objects and vertical businesses like earlier generations. The potential size of mobile subscriptions has grown from billions of people to practically innumerable interconnected persons, computers, and things. It supports services like mobile broadband, Industry 4.0, Virtual Reality (VR), Augmented Reality (AR), IoT, and automated driving (Hakeem et al., 2020). Online schooling, remote working, automated unmanned vehicles, robotics, and intelligent healthcare represent the 5G services that demonstrate the relevance of networks digital infrastructure. However, 5G is currently being implemented globally; it is time for academics and industry to focus on 6G networks to meet the future of Information Communication Technol-

ogy (ICT) needs in 2030. We introduce Some pioneering efforts on next-generation wireless networks that have been undertaken.

In July 2018, the ITU-T standards sector created a focus group named Technologies for Network 2030.

The organization plans to research network capabilities for 2030 and beyond (Rout, 2020), including holographic communications, intelligence, multi-sense experience, Tactile Internet, and digital twin. Many projects for the necessary 6G technologies were launched at the start of 2020. Intending to shape Europe's digital future, the European Commission accelerate investments in Giga-bit Connectivity project (Chowdhury et al., 2020). Next Generation Mobile Networks (NGMN) announced its 6G vision in October 2020. ITU-R agreed in February 2020 to initiate research on future technological developments for the evolution of International Mobile Telecommunications (IMT) (Series, 2015). Its 6Genesis program focuses on challenging research areas like reliable unlimited wireless connectivity, distributed computing, intelligence, and antennas for future circuits and devices (Chen et al., 2020). Several prominent US actors in mobile communications have also started or announced 6G research initiatives.

Unprecedented growth in smart products, services, and applications necessitates the rapid expansion of mobile communications infrastructure. The expansion of mobile broadband has exponentially increased the number of smartphones and tablets over the previous decade Mobile broadband (MBB). Mobile device adoption is still low in undeveloped nations; thus, this trend will continue until the 2020s. Fig. 2 shows that by 2030, 17.1 billion people will be using MBB. However, as the number of MBB users grows, the traffic demand per MBB user grows. This is due to the popularity of mobile video services and the constant development in mobile screen quality.

Machine-to-Machine M2M terminals will become saturated in 2030, along with human-centric communications. Global M2M subscribers will reach 97 billion by 2020, and 14 times increase as shown in Fig. 3.

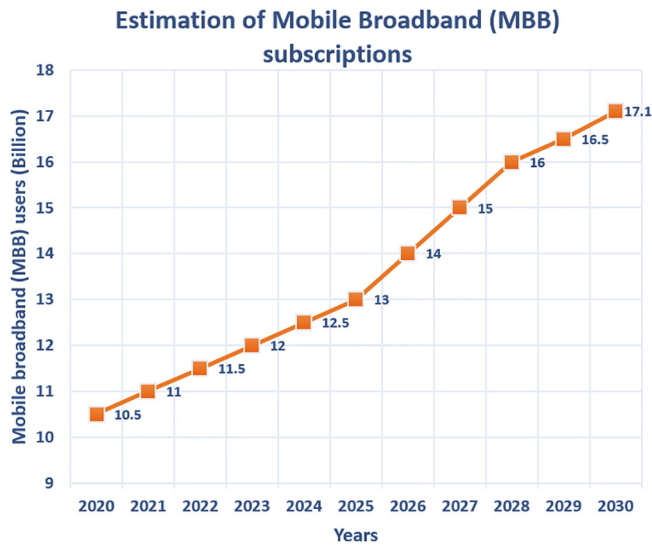


Fig. 2. Estimated mobile broadband users subscriptions according to ITU-R Report M.2370-0 from 2020 to 2030 in Billion.

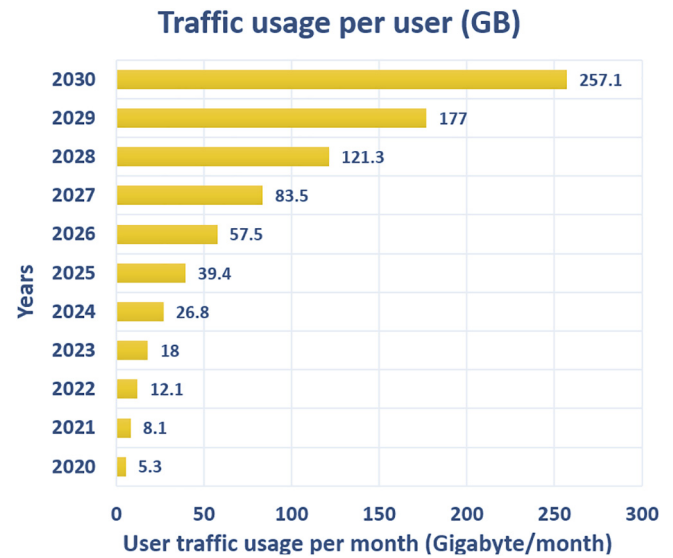


Fig. 4. Estimated user traffic per month according to ITU-R Report M.2370-0 from 2020 to 2030 (1 exabyte (EB) = 10^6 terabytes (TB), 1 TB = 10^3 gigabytes (GB)).

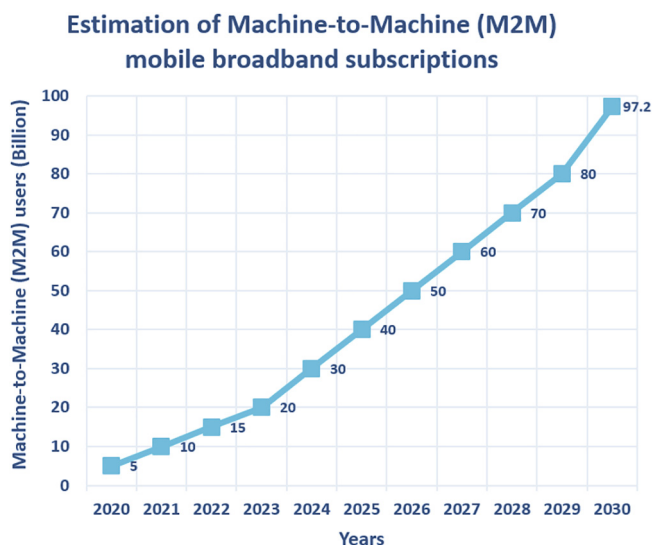


Fig. 3. Estimated Machine-to-Machine (M2M) broadband subscriptions according to ITU-R Report M.2370-0 from 2020 to 2030 in Billion.

A recent study found that mobile video traffic already accounts for two-thirds of all mobile traffic (Zhang et al., 2019). Rich-video services are driving traffic growth in certain developed nations and AR and VR applications will continue to expand traffic long-term in these countries. Monthly mobile data usage will rise from roughly 5 GB in 2020 to over 250 GB in 2030, as shown in Fig. 4.

The massive growth of mobile traffic in 2030 and beyond will strain the 5G system. According to ITU-R, global mobile traffic will grow from 62 EB per month in 2020 to more than 5036 EB per month in 2030 due to many rich-video applications, higher screen resolution, and mobile cloud services (Union, 2015). This is supported by a recent Ericsson report, which shows that global mobile traffic reached 33 EB per month in late 2019 (Ericsson, 2020), as shown in Fig. 5.

1.2. Paper contributions

This paper included the 6G drivers and detailed traffic growth according to the standards and recent industry reports and white

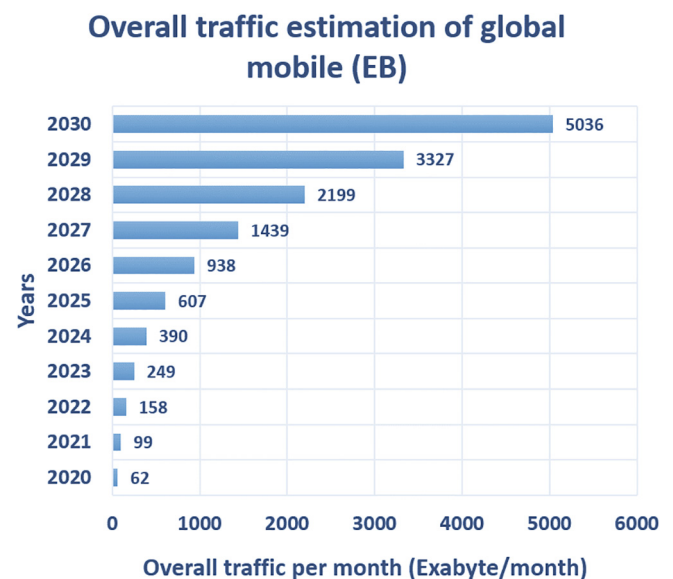


Fig. 5. Estimated overall mobile traffic per month according to ITU-R Report M.2370-0 from 2020 to 2030 (1 exabyte (EB) = 10^6 terabytes (TB), 1 TB = 10^3 gigabytes (GB)).

papers. We are proposing a road map for the future of 6G based on the current research efforts and current 6G projects. Most of the 6G surveys and tutorials focus on many aspects without providing a detailed framework structure for the mentioned technologies. Also, it lacks the industrial sector's contribution, which is very important for any technology to be deployed and implemented. A few surveys discussed the industrial application sub-sectors of 6G (Nguyen et al., 2021; Padhi and Charrua-Santos, 2021; Padhi and Charrua-Santos, 2021). They encourage us to include a comprehensive discussion of the industrial projects, their funds, results, and plans concerning implementing and deploying new 6G industrial applications. This paper covers the most significant directions of 6G networks. We also provide a detailed description of some emerging applications, their benefits, concepts, challenges, and requirements. We summarize the paper contributions as follows:

- Studying the communication difficulties introduced into the early legacy mobile networks.
- Discussing how the advances in 5G design impact the new architecture of 6G
- Discussing the evolution and timeline of advanced 5G
- Proposing the 6G technological framework components and their requirements
- Presenting the 6G network architectural vision and related problems
- Introducing the hardware-software evolution from a 6G perspective
- Studying the recent beyond 5G research activities in terms of contributions, funding, and outputs
- Presenting the latest 6G projects, stating their challenges, coordinators, and participants.
- Proposing an expected framework for the 6G applications and their requirements
- Presenting the latest 6G applications and their needs

The rest of this paper is organized as follows. [Section 2](#) shows the development of mobile cellular networks, the evolution of 5G, and the new scenarios beyond 5G. [Section 3](#) introduces the 6G current related work. [Section 4](#) introduces 6G requirements and capabilities. 6G network architectural vision and hardware-software evolution are discussed in [Section 5](#). [Section 6](#) provides B5G and 6G vertical industries projects. [Section 7](#) introduces the leading expected applications of 6G, while the study is concluded in [Section 8](#). [Table 1](#) summarizes the acronyms used in this paper.

2. Evolution of earlier mobile networks

This section discusses the evolution of different wireless mobile generations starting from 1G to 6G. We also present the evolution

Table 1
Summary of acronyms.

Acronyms	Definition
mmWave	millimeter-wave
THz	Terahertz
IoE	The Internet of Everything
EURLLC	Enhanced Ultra-Reliable and Low Latency Communication
AMPS	Advanced Mobile Phone System
NMT	Nordic Mobile Telephone
GSM	Global System for Mobile Communications
CDMA	Code-Division Multiple Access
LTE	Long Term Evolution
VR	Virtual Reality
AR	Augmented Reality
ICT	Information Communication Technology
NGMN	Next Generation Mobile Networks
IMT	International Mobile Telecommunications
MBB	Mobile Broadband
M2M	Machine-to-Machine
FCC	Federal Communications Commission
MIMO	Massive Multiple-Input, Multiple-Output
3GPP	The 3rd Generation Partnership Project
IAB	Integrated-Access Backhauling
CN	Core Networks
RAN	Radio Access Network
OWC	Optical Wireless Communications
DRL	Deep Reinforcement-Learning
IWD	Intelligent Wearable Devices
H2H	Hospital-to-Home service
WBANs	Wireless Body Area Networks
UAVs	Unmanned Aerial Vehicles
FAA	Federal Aviation Administration
LoS	Line-of-Sight
SLAM	Simultaneous Localization and Mapping
CAV	Connected Autonomous Vehicles

of 6G new scenarios and use cases according to the demands of future applications. The evolution of cellular mobile communications is shown in [Fig. 6](#).

2.1. Evolution from 1G to 4G

Initially, the 1G network was designed only to provide voice communications in the 1980s. It uses analog modulation techniques to transport data and does not comply with a defined wireless standard. This generation has several issues, including handover, security, and transmission. In the second generation of mobile phones, digital modulation technologies like TDMA are used to deliver voice and text messaging services. This era relies heavily on the GSM mobile communication system. The challenge and response technique is used in 2G authentication. The 3G network has provided customers with download and uploads rates of up to 2Mbps since its launch in 2000. Modern services such as TV streaming, internet browsing, and video streaming are now accessible at a rate previously unimaginable in mobile communication. Up to 1 Gbit/s transmission rates were achieved in down-link transmission on 4G networks in 2009 ([Chih-Lin et al., 2016](#)). 4G networks can handle high spectrum efficiency and low latency (DVB). Many intelligent mobile terminals are included in 4G networks, IP core networks, backbone networks, and access networks. For 4G networks, authentication and tampering are two of the most critical security issues.

2.2. Evolution of 5G

As the 5G network nears commercialization, complex systems and high-security designs may increase data speeds. Unique to 5G is its potential to connect ever-increasing devices while still offering higher-quality services to all network participants.

2.2.1. 5G timeline

In 2012, the ITU communication technology organization began transitioning to 5G by launching a program to create IMT systems for 2020 and beyond. According to GlobalData, the following are the most significant milestones in the 5G theme's journey.

In 2012, the International Telecommunication Union (ITU-R) began working on a program to establish Internal Mobile Telecommunications (IMT) networks. UK's University of Surrey revealed plans for a 5G innovation center in 2012. The European Commission funded 5G networks as part of its Horizon 2020 initiative, launched in 2013. ITU-R released its vision for IMT-2020 in 2015, which included a list of 5G needs.

Pre-commercial testing of Verizon's proprietary 5G radio standard was announced in 2016.

The European Union (EU) encouraged public and private investment in 5G infrastructure. The High-band spectrum for 5G wireless services was granted by the US Federal Communications Commission (FCC) in 2016. 3GPP approved the first 5G release in 2017, a significant step towards constructing commercial 5G networks.

In the second part of 2017, China's three leading operators tested 5G technology in many cities. In 2018, the Pyeongchang Winter Olympics included 5G application demos. The 5G spectrum auction in South Korea was concluded, and the deployment is scheduled for April 2019.

3GPP agreed on 5G standards that may be used in areas with no existing infrastructure, paving the way for 5G deployment. As of 2018, mobile carriers in the United Kingdom have invested £1.4 billion for 2.3 GHz and 3.44 GHz spectrum. New mobile 5G service from AT&T was available in a dozen major metropolitan areas throughout the United States in 2018. ZTE unveiled the ZTE Axon 10 Pro 5G in China, the Nordic nations, and the Middle East in 2019 as the first 5G-ready mobile smartphone.

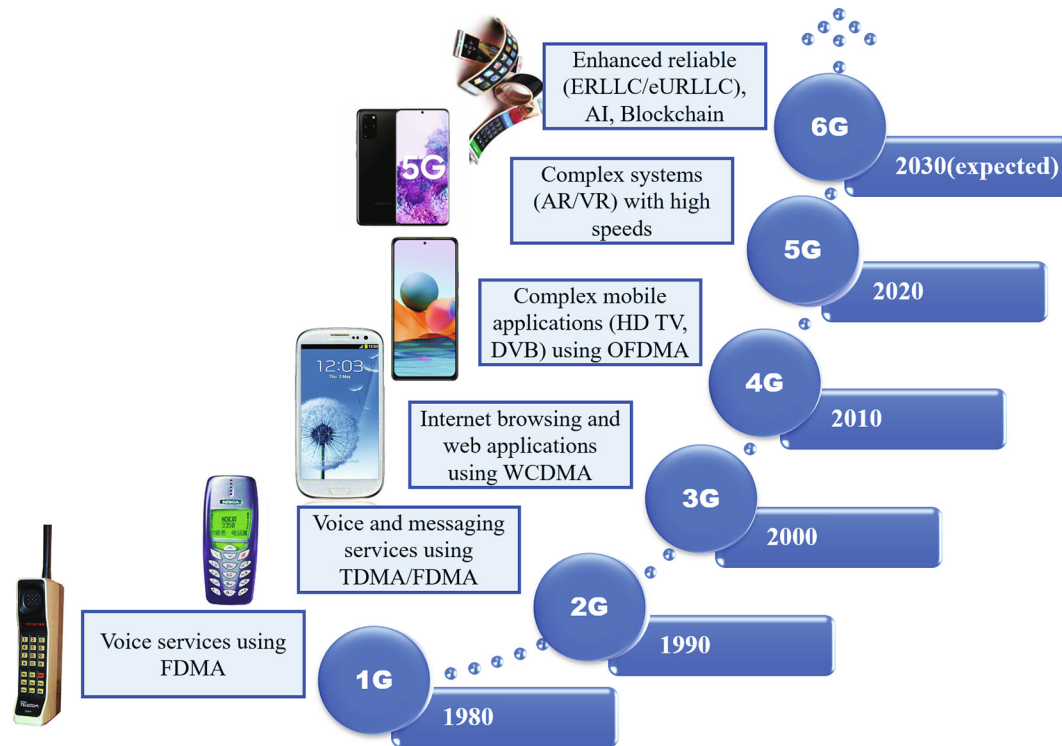


Fig. 6. The evolution of cellular mobile communications.

Release 16 (NR Phase 2) of the 5G standard was completed in 2020. As of July, China has over 100 million 5G customers, making it the world's most popular 5G network. The COVID-19 pandemic delayed the completion of 3GPP Release 17 standards in 2020.

Japan plans to employ 5G technology for smart city applications during the postponed Tokyo Summer Olympic Games in 2021. Release 17 standards are expected to be completed in 2022. There will be three billion 5G mobile customers in 2025, or 28 percent of all mobile subscribers, according to GlobalData. In 2030, the so-called 6G technology will have matured (Chandramouli et al., 2019).

2.2.2. 5G features, deployment modes, and plans

Release 18 and subsequent releases of the next 5G cellular standard will be known as 5G Advanced when it was announced by the 3rd Generation Partnership Project (3GPP) in April 2021. The non-standalone (NSA) new radio (NR) standards for 5G technology were delivered in late 2017. Both NSA and non-NSA 5G networks are being deployed throughout the globe as the first phase of 5G deployment begins. Awaiting the advent of the 5G Advanced standard now, the cellular sector is expecting its arrival. 3GPP has announced that the current standard would be substantially upgraded to increase efficiency and add features.

Nokia's Bell Labs claimed that 5G Advanced offers high capabilities new services, extreme connection, and high security (Weldon, 2016). Enhanced massively Massive multiple-input, multiple-output (MIMO) performance in a 5G network is improved with MIMO 5G Advanced. Data transmission rates may be increased when numerous antennas are used at both the transmitter and receiver. Increasing the number of transmitters and receivers in a system to support many users is known as massive MIMO. According to Bell Labs, base stations that handle 64 to 512 transceivers are expected to be required for 5G Advanced and future standards. As a result, 5G Advanced systems for business networks will be able to

serve networks with many high-throughput data users better. We discuss the achieved features and plans of the advanced 5G below.

- **Reducing capacity:** RedCap (Reduced Capacity) will be part of the standard for Release 17. An upgrade to 3GPP's Release 17 specification called RedCap specifies how 5G-capable devices use less bandwidth under the new standard. RedCap allows wearable and IoT devices to function in a 20 MHz channel instead of the 100 MHz channel generally used by 5G NR. Reliability with current 5G NR and RedCap devices will be maintained while the bandwidth requirements for RedCap devices are reduced in Release 18.
- **Accurate timing:** Many industries, such as banking, industrial IoT, and smart energy grid technologies, depend on millisecond precision timing. Nokia said 5G Advanced would protect satellite communications from deterioration or loss by supplying time references straight from the network. Enterprise users will be impacted by the 3GPP Release 17 upgrade freeze, which is expected to last until the second quarter of 2022. Compatible devices will be able to communicate with 5G satellites after a non-terrestrial network upgrade. Wearables and IoT devices will benefit from a slower transmission rate in Release 17.
- **Improved positioning:** Customers will benefit from improved location accuracy due to the revised standard. Using beamforming and time-based approaches, Nokia claims 5G Advanced will increase cellular-positioning precision to sub 10 cm. In the absence of a GPS satellite connection, devices will be able to move with centimeter precision inside.
- **5G advances using AI:** Advances in AI and ML will be used in 5G Advanced to enhance the performance of multi-antenna systems and implement intelligent network management. Extended reality (XR) communication, which builds on proven AR and VR applications, may be supported by this technology. Improved energy efficiency is a fundamental goal of 5G

Advanced. According to Ericsson, this will be achievable with the network's AI and ML upgrades. Release 18 of the 3GPP specification will include a model for base station energy usage and ways for reducing power use.

- **5G Operators Advances:** However, carriers are already planning for a more advanced version of 5G. From XR experiences to AI and ML approaches in the network, AT&T Labs' claimed that Release 18 would offer a variety of system upgrades, including support for vertical services. The report mentioned that these requirements will not be completed until early 2024. AT&T is participating in the 3GPP and work closely with the requirements. It anticipates the features of Release 18 to be implemented within 18 to 24 months. 5G current phase is expected to provide significant improvements in data throughput, location, time, and coverage (Agiwal et al., 2016; Ziegler et al., 2019). All of the improvements paved the way for the evolution of 6G.

2.3. Evolution of 6G

The New 6G applications will have more requirements and a greater network capacity than the current 5G networks. As a result, new 6G applications will need a more extensive network capacity than 5G networks. Next-generation wireless networks will be one of the significant components in our future lifestyles, industries, and societies. Wireless networks will be the link between humans and intelligent machines. Therefore, the research community and industry should improve these networks towards a shared vision (Goyal et al., 2019). The 2030 era will witness a considerable improvement in wireless communication. Future communication should have fundamental divers: systems trustworthiness; sustainability of devices efficiency, automatization and digitalization for a simplified life, and limitless connectivity to satisfy application demands. It is expected that this era will have a great transformation towards automatization, where 6G will play a vital role as a communication and information backbone. 6G should allow anything to communicate anywhere and anytime.

Nowadays, the next generation of wireless networks should achieve several factors to meet these applications' QoS due to new applications requirements. Furthermore, on the road to 2030, we can presume the next evolution to go on with higher

needs that the wireless networks should support. Next generations should support extremely high data rate, very low end-to-end delay, more reliability, massive cell-capacity, and extended coverage area to support highly demanding applications such as virtual, augmented, and mixed reality and remote control of sensitive operations. The upcoming generation should achieve the following requirements (Jaber et al., 2016; Khan et al., 2020; Tomkos et al., 2020; Hayes, 2020).

- Very low latency communication (10 μ s–100 μ s).
- The peak data rate is at least 1 Tb/s.
- Support high mobility (≥ 1000 km/h) to support airline systems.
- Huge capacity (10^7 device/km²) due to the IoT.
- Reduce energy consumption to introduce green communication (100 $\times$ reduction compared with 5G)
- Data rate reaches 10 Gb/s (*10 times 5G).
- Spectrum efficiency (5–10 times 5G).
- High 3D coverage extension including terrestrial, aerial, space, and sea domain.

Fig. 7 shows the quantitative comparison between 5G and 6G in terms of basic requirements in each network.

2.4. New scenarios of 6G beyond 2030

New application scenarios will continue to rise until 2030. The scenarios are divided into three categories: intelligent production, intelligent life, and intelligent society. Typical use case after 2030 (Sodhro et al., 2020; Khan et al., 2020).

- **Smart Production:** The digital economy might overgrow by applying developing technology to agriculture and industry. 6G will achieve intelligent manufacturing via informatization. For example, drones are utilized in agriculture. Robotics and virtual reality will boost production efficiency. With modern technologies like digital twins, 6G will have more significant intelligent manufacturing.
- **Smart Life:** Twin body area network Synesthesia internet and intelligent interaction will likely transform our lives in 2030.
- **Smart Society:** The ubiquitous coverage network in 2030 will considerably extend public service coverage, bridging the digi-

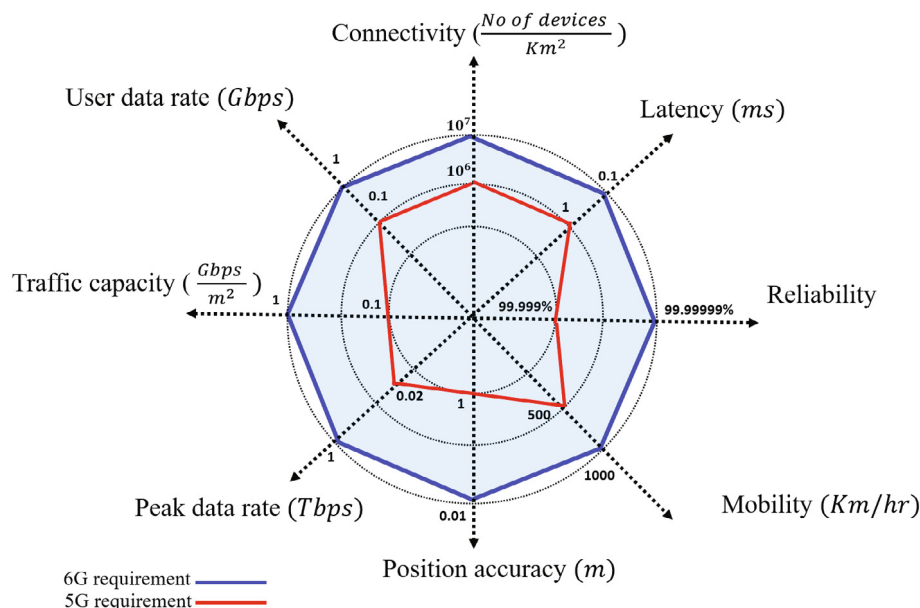


Fig. 7. Quantitatively comparison between the 5G and future 6G in terms of requirements.

tal divide between areas. Overall, a 6G network will strengthen social governance and provide a firm basis for a better society. Fig. 8 shows the new scenarios arising in 6G networks. We also introduce some use cases in the 5G networks and 6G networks and compare them in terms of key requirements, as shown in Table 2.

3. Existing 6G surveys

6G networks have been the subject of numerous studies (Yang et al., 2019; Letaief et al., 2019; Tariq et al., 2020; Zong et al., 2019; Tang et al., 2019; Dang et al., 2019; Viswanathan and Mogensen, 2020; Zhang et al., 2019; Giordani et al., 2020; Chen

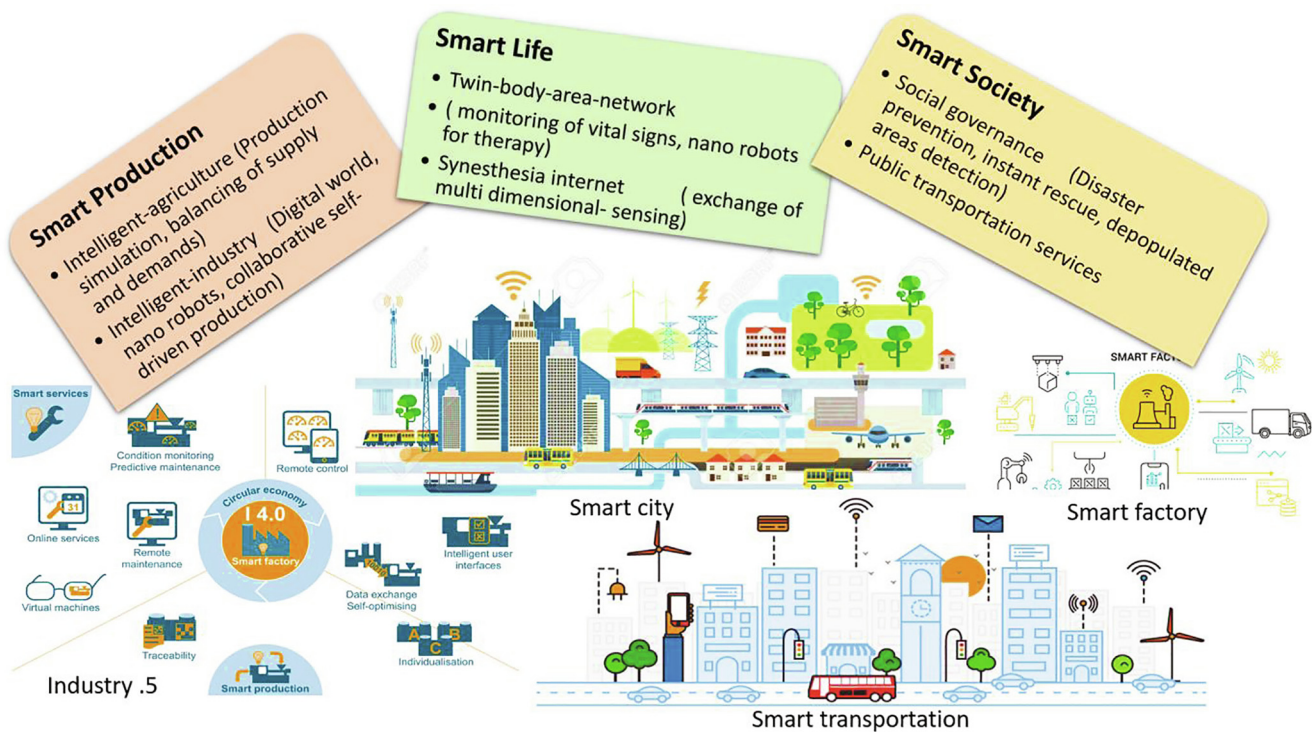


Fig. 8. Three scenarios of 6G beyond 2030.

Table 2
5G vs. 6G: Requirements and their uses cases.

	5G	Use Cases	6G	Use Cases
Data Rate	1 – 10 Gb/s	Telemedicine	100 Gb/s – 1 Tb/s	3D holographic AR/VR Robotics Arm
Coverage Extension	0.1 km	The limited scale of IoT network	3D coverage scenarios (10000 m (Sky, 200NM (sea)	Terrestrial, aerial, space and sea domain, massive-scale IoT network
Power Consumption	10 years battery life	IoT devices	50 times improvements compared to 5G, nearly (1 Tb/J)	Wearable user devices Zero energy devices
End-to-End Latency	1–5 ms	Vehicular Networks Military Services	< 1 ms	Healthcare Networks AR/VR Unmanned Ariel Vehicle (UAV) Robotics Arm
Reliable Communication	99.9%	Vehicular Networks Telemedicine	~ 99.9999%	Healthcare Networks AR/VR Unmanned Ariel Vehicle (UAV) Robotics Arm
Massive Connectivity and Sensing	1 milliondevice / km ²	IoT devices	10 milliondevice / km ²	Wearable user devices AR/VR IoE
Frequency Extension and Improved Spectrum	3 – 300 GHz	mmWave for fixed access	Up to 1 THz	mmWave Sub-6 GHzExploration of THz bands (above 300 GHz) high-definition imaging and frequencyNon-RF (e.g., optical, VLC) spectroscopy localization
Mobility and speed supportive	500 km/hr	Vehicular Networks	1000 km/hr	Terrestrial, space, sea, aerial, and airline

et al., 2020; Saad et al., 2019). Recently, there has been a noticeable increase in the number of publications linked to 6G. Yang et al. (2019) looked into 6G wireless network technology options. The authors first offered an outline of 6G's time–frequency-space resource consumption. Letaief et al. (2019) described the wide range of AI-enabled innovative services in 6G cellular systems. They also studied the 6G architecture based on AI and the emerging AI applications. Tariq et al., 2020) introduced the 6G applications, the supporting technologies, and open research problems. Zong et al. (2019) surveyed the 6G technologies and their architecture. Tang et al. (2019) reviewed several machine learning technologies using vehicle networks and presented a roadmap to an intelligent 6G network.

In (Dang et al., 2019), Dang et al. showed that the 6G objective should be human-centric, and security and privacy are the essential elements to support 6G. They also predict that integrated frameworks and new technologies must achieve this objective.

According to Viswanathan and Mogensen, new topics that are expected to define 6G demands have been outlined in their study of communication demands and technology in 6G (Viswanathan and Mogensen, 2020). Zhang et al. (2019) predicted that the 6G success secret is a rapid and significant cost reduction from the customer's perspective.

The authors also discussed possible future problems with the rollout of the 6G network. They focused on 6G's core technology and future vision. 6G use cases and requirements were discussed regarding maximum data rate, application data rate, over-air latencies, energy consumption, and connection density. Giordani et al.

(2020) focused on the wireless communication technologies that evolved towards 6G and their use cases. Also, the authors focused on 6G major enabling technologies and their accompanying problems and potential uses. In addition, they talked about how 6G networks will incorporate intelligence.

Chen et al. (2020) provided a comprehensive review that aims to shed light on how 6G mobile communication systems may address the issues of throughput, data rate, coverage, and mobility. Saad et al. (2019) highlighted the 6G implementations, technologies, and unaddressed research problems. Kato et al. (2020) examined several machine learning approaches and identified many critical challenges in intelligent 6G systems.

Gui et al. (2020) defined AI for 6G covering 6G standardization, performance indicators, methodologies, and a framework for the future wireless systems.

Tataria et al. (2021) proposed a new comprehensive discussion of the 6G challenges and opportunities. This is followed by a discussion of the technical requirements required to allow 6G applications. They examined significant difficulties and opportunities for realistically realizable system solutions across all layers (i.e., from applications to the physical layer).

Mahdi et al. (2021) studied the technological constraints of 5G wireless networks and the projected obstacles of 6G communication networks. They looked at the applications of three extremely demanding sectors: Internet-of-Things, energy (IoT), and machine learning.

Hakeem et al. (2022) proposed a discussion of 6G applications and technologies and introduced the 6G network's visions and

Table 3
Existing surveys and tutorials comparison.

State-of-art	Discussed topics	Major Contributions	Publishing date
Yang et al. (2019)	Tutorial about 6G enabling drivers	- Offering an outline of 6G's time–frequency-space resource consumption. 6G enabling drivers	July 2019
Letaief et al. (2019)	6G vision	- Introducing 6G vision - Presenting 6G network architecture and key enablers in 6G	August 2019
Tariq et al. (2020)	Tutorial about 6G technologies and open issues	Introducing the 6G applications, Supporting technologies, and open research problems	August 2019
Zong et al. (2019)	The tutorial discusses the 6G key drivers	- Studying the 6G key drivers and technologies - Studying the 6G new architectures requirements - Introducing the possible future problems of the 6G network	September 2019
Tang et al. (2019)	Machine learning in vehicular communication based on 6G	- Reviewing of several machine learning technologies using vehicle networks - Presenting a roadmap to an intelligent 6G network	December 2019
Dang et al. (2019)	6G vision and security challenges	6G objective is human-centric, Security and privacy are the essential elements to support 6G. Integrated frameworks to achieve 6G objectives.	January 2020
Viswanathan and Mogensen (2020)	6G survey regarding new demands	- Presenting new technologies, new spectrum, new security solutions - Study the communication demands and the AI usage in 6G	March 2020
Zhang et al. (2019)	6G survey regarding assisting AI in 6G	- Reduction of 6G costs using AI applications - Studying the customer's needs in 6G networks	March 2020
Giordani et al. (2020)	Tutorial to present the primary 6G technologies	Reviewing the current wireless communication technologies towards 6G and their use cases	March 2020
Chen et al. (2020)	Survey about 6G issues	Addressing the issues of throughput, data rate, coverage, and mobility	April 2020
Saad et al. (2019)	Tutorial about 6G technologies	- Presenting open research problems - Enabling 6G technologies	May 2020
Kato et al. (2020)	Tutorial regards deploying machine learning in 6G	- Examining several machine learning approaches - Identified critical challenges in intelligent 6G	June 2020
Gui et al. (2020)	Survey about AI for 6G	AI for 6G covering 6G standardization, performance indicators, methodologies, and a framework for future wireless systems.	June 2020
Tataria et al. (2021)	comprehensive survey about the discussion of the 6G challenges and opportunities	Discussing the technical requirements required to allow 6G applications	March 2021
Mahdi et al. (2021)	Survey about 5G to 6G improvements	Looking at the applications of three extremely demanding sectors: Internet-of-Things, energy (IoT), and machine learning	August 2021
Hakeem et al. (2022)	Survey about 6G applications and technologies in terms of security aspects	- Studying the 6G security challenges and requirements - Comparing the previous cellular legacy mobile networks in terms of requirements and security issues - Proposing 6G security structure - Studying the applications of 6G	March 2022

structure. They also proposed the security structure and requirements of the promising 6G technologies and applications. Moreover, they discussed all legacy mobile networks before 6G in a detailed manner to explain the obstacles and improvements that affect the deployment and implementation of 6G networks.

Table 3 shows the main discussed topics and the significant recent contributions related to 6G.

4. 6G requirements and capabilities

This section introduces the requirements and needed capabilities to deploy a 6G network (Wikström et al., 2020), as shown in Fig. 9. Fig. 9 also clarifies the leading technologies needed for 6G networks enhancement.

4.1. Network flexibility

6G targets to enhance the energy usage, cost of installations, network growth, and administration to improve network flexibility. Mechanisms for ensuring active network installations will be critical in the future for deploying a low-cost and resilient network with high capacity. The main difficulty is combining conventional service providers of the deployed network nodes with non-terrestrial, ad-hoc, mobile nodes. The 5G Integrated-Access Backhauling (IAB) and multi-hop communication will be vital for the dynamic 6G network deployments. There is a need for enhancing the transport layer to be flexible, scalable, and reliable to handle the 6G use cases' new deployment choices (Song et al., 2020).

Future deployments will be less centralized with more shared platforms for core networks (CN) and Radio Access Network (RAN). In addition, they are eliminating the duplicate of network functions by combining RAN and CN functions. The appropriate combination is required to offer new use cases with higher perfor-

mance and deployment flexibility while keeping network operations and development efforts manageable (Tomkos et al., 2020).

4.2. End-to-end communication

Future 6G services require high-performance connections to meet the dynamic behaviors, most extraordinary bandwidth requirements, and robustness. In addition, end-to-end network connectivity impacts resilience measures, network cooperation, end-to-end evolved protocols, and latency management. Network applications that need to be resilient in end-to-end transmission and connection must be supported. Similarly, the critical network infrastructure must be accessible and have commercial monitoring. Furthermore, the increasing need to secure end-to-end communication requires excellent collaboration between all network entities and applications. Future communications utilize new restricted requirements for multi-access technologies and different applications. This is a significant advance in developing 6G networks to manage resilience and traffic (Han et al., 2021). Many services and use cases require lower latency requirements with maximum latency tolerance. In addition, predictable latency will enable new use cases with centralized and distributed deployment methods.

4.3. High coverage

The new radio access solutions should deliver excellent performance in several ways to allow new services at affordable prices. First, higher data rates and latency performance are needed. Moreover, excellent network access coverage is necessary to keep higher service requirements. Therefore, 6G requires introducing novel mesh networking technologies and more integrated access and backhaul networks to enable increased capacity and dense network deployments at a low cost (Giordani and Zorzi, 2020).

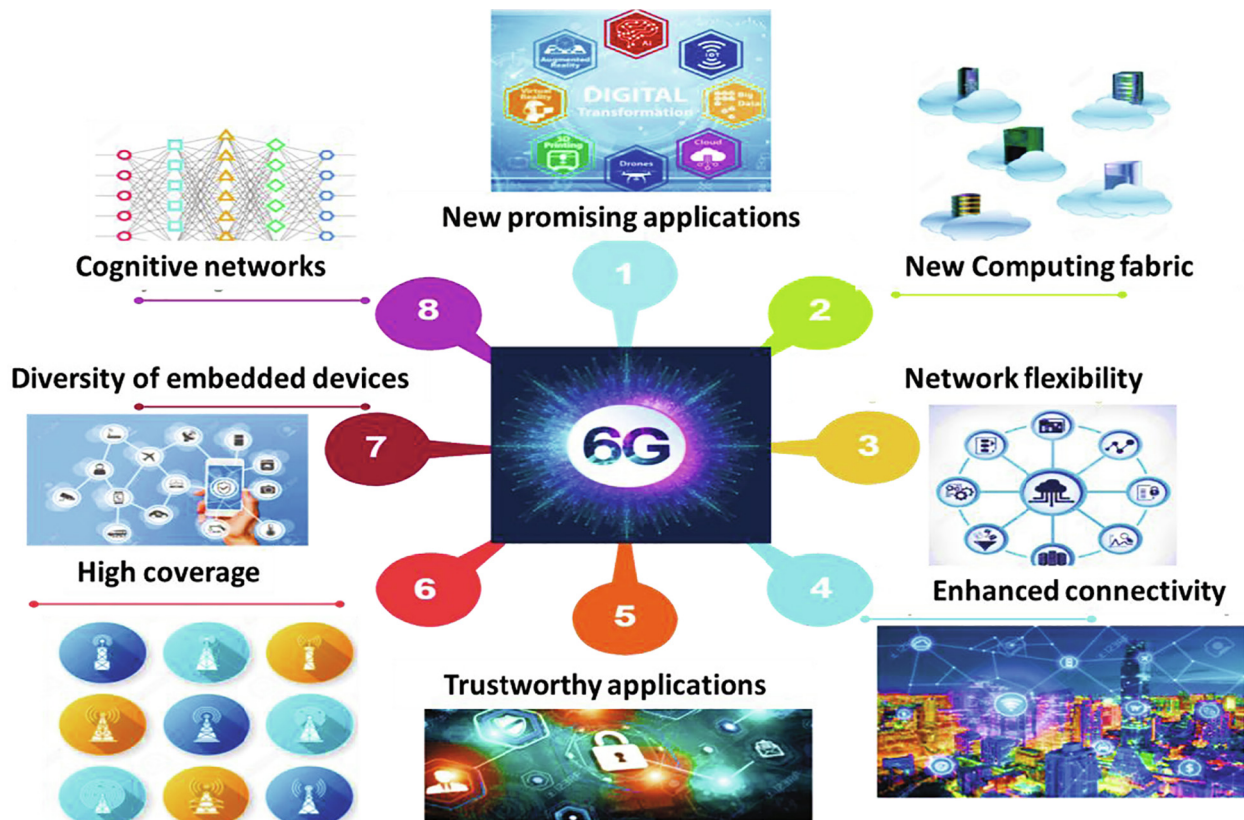


Fig. 9. 6G capabilities and key technological components.

For a wireless connection, the spectrum is a vital resource. Low-frequency bands of 6 GHz are utilized by 4G/5G, although they will also be significant in the 6G era (Liu et al., 2020). Given that minimal new spectrum is expected, 6G radio access must share spectrum with earlier generations. 6G will utilize the mmWave bands between 24 and 52 GHz founded by 5G and scheduled to be expanded up to 100 GHz. There are possibilities for reasonably significant quantities of spectrum beyond 100 GHz. 6G will expand network access to traditional terrestrial and non-terrestrial radio access. Drones and satellites can represent non-terrestrial access networks. It will be a seamless component of the wireless access system, offering global coverage (Vaezi et al., 2021).

4.4. Diversity of embedded devices

Future services will need connection in all places and situations. 6G networks can handle billions of tiny embedded devices while providing reliable, always-on connections (Hu et al., 2020). High machine-to-machine communication now offers data speeds of up to 100 kbps. However, their battery life may last up to 10 years in certain situations, the devices' value is limited by battery replacement or charge. However, highly energy-efficient communication protocols must be devised since the quantity of energy harvested is usually relatively tiny (Qi et al., 2020).

4.5. Cognitive networks

We must comprehensively reuse the degree of intelligence of these networks to achieve expected networks with lower costs. Cognitive networks will aid in the improvement of energy efficiency and the availability of services. We suppose this to happen in two ways: in difficult-to-achieve optimizations using conventional methods, where ML and AI can help, and in developing controls systems to perform system management functions autonomously (Zhang et al., 2020). By expressing operational objectives in the form of intentions, humans will influence what systems perform. Automated management necessitates a greater degree of abstraction in the user-machine interfaces and the capacity of the strategies to understand and reason about such objectives. Humans and analytics algorithms will contribute knowledge and experience presented in shared learning. A cognitive network (Hussein et al., 2013) would then utilize these components to include various circumstances, identify appropriate corrective actions, and plan the optimal ways of action for their performance in the network (Wei et al., 2020).

Cognitive systems must adapt to their surroundings naturally, continuously monitoring and learning from previous events. Knowing of earlier operations performance are sent back to enhance settings, procedures, and software in real-time. We will see continual improvements against attacks spread across physical and logical functions inside a network. This continual optimization can considerably make the complete system more securely dynamic than today. Intelligence will be accessible in many ways over a distributed network (Yang et al., 2021).

4.6. Network computing

6G will bring all physical devices into the computation. It will control physical systems. By virtualizing computing and storage, service providers can improve application performance, reliability, and latencies. Moreover, network computing will provide services and tools with connections for businesses and industries (Du et al., 2020). To communicate with physical systems, new applications are developed. As a result, computing faces new challenges. New positioning, arrangement, and software programming methods are required to meet real-time deadlines. It will also be necessary

to expose and maximize the use of energy-efficient, specialist computing equipment. Conversely, application development must be more superficial. The ecosystem will need to innovate once more objects are linked. More effect is required to meet the growing demand for customized applications. Besides standard APIs and abstractions, new programming ideas and simpler models are included. Network computing facilitates the creation of new software and hardware systems. The network edge, central cloud, and devices will all get apps. As a result, 6G will be a true innovation hub.

4.7. Trusted systems

A foundation of building reliable systems is the capability to resist, detect, react to, and recover from attacks and incidental confusions. The four essential building elements for trustworthy systems are using private computing solutions, secure identifications and protocols, service availability, and security insurance and defense. AI is supposed to significantly affect future technological development and security in the four domains above. Concurrently, the trustworthiness of AI components becomes critical (Ji et al., 2021). Security support and certification are obtaining much traction these days. Current state-of-the-art security support designs are effective methods for ensuring the security of a unique product version. However, more improvements are needed. Virtualization and cloud computing enhancements, as well as continuous integration and continuous delivery procedures, and AI, must all be considered. Today's security preserving methods should be updated in the future to evaluate all system elements. It is critical to create well-defined criteria and procedures approved by all stakeholders when developing security schemes.

Confidential computing is a powerful paradigm for protecting data during processing and storage. Encryption is used to protect data from being tampered with within the cloud. The hardware handles the certification and processes, preventing the cloud provider from changing or altering the hardware. The Root-of-Trust (RoT) mechanism is fundamental to secure computing. Secure protocols can be achieved by combining data protection methods with private computing and storage and connecting devices, infrastructure, and network slicing tasks. It was identifying physical components and software functions using RoT. We target to create a system that protects the data while ensuring application privacy. Resource provisioning for key services is structured to enable varying service guarantees. Automatic recovery mechanisms that analyze and aggregate data from all communication system sections are also available. A distributed and hierarchical strategy is required for better performance observability and intermediate analytics to verify that criteria are met in real-time. AI can also be used to validate end-to-end service availability using data-driven observability. Real-time AI analytics may also improve resilience to changes in traffic load and radio settings (Stergiou et al., 2020; Kantola, 2020).

5. 6G network architectural vision and hardware -Software evolution

This section discusses the 6G network architectural visions and the future hardware components improvements. We study the hardware-software complexity and requirements to support the 6G high requirement services with higher data rates and low latency communications. We discuss how the current HW-SW designs and approaches are evaluated at a low cost to enhance the future of 6G networks.

5.1. 6G architectural vision

Communication components like base stations and terminals may conduct ubiquitous sensing, which is the basis for connecting all intelligent devices on an intelligent platform. ICT and vertical industry ecosystems, create more excellent stability, sustainability, and health. As a consequence, the whole civilization might be digitalized. So, from an architectural standpoint, we define the 6G vision as follows (Khan et al., 2021).

- **Connected Intelligent network:** 6G is the next-generation mobile communications technology that will be a distributed neural network to connect the physical and virtual worlds, resulting in an age of sensed, linked, and intelligent networks. This massive neural network's edge communicates with other neural edges or neural centers autonomously and seamlessly. The entire connectivity between 6G network edges supports the future Intelligence of Everything (Manogaran et al., 2021).
- **Integration of AI and ML:** 6G air interface and network designs will use AI/ML to automate system operation and administration. Decentralized intelligence in the cloud gives way to ubiquitous intelligence at the edges as each 6G network integrates computation, communication, and sensing capabilities. A distributed machine learning architecture will be essential for society and vertical businesses. 6G promises a ubiquitous intelligent mobile communication system that redefines terminal, cloud boundaries, and connections.
- **The networked sensing:** Communications systems will use wireless sensing capabilities to examine the physical environment via radio wave transmission, reflection, and scattering. They will also increase communication performance and enable more comprehensive network service situations, creating the data basis for constructing an intelligent digital world.
- **Full coverage connectivity:** 6G will provide speeds equivalent to optical fibers. 6G achieves ten times increase in 5G connection density, millimeter-level imaging, centimeter-level localization, and end-to-end reliability. It enables future human-centric services and accelerates full-scale digital transformation and upgrade of vertical industries.
- **Integration of terrestrial and non-terrestrial networks:** Many satellites will be placed in non-terrestrial networks to construct a mega satellite constellation. Low-latency ultra-long-haul transmission will increase the reach of terrestrial cellular infrastructure. Both networks are anticipated to be extensively linked as one system, allowing users to use the benefits of each kind under varied service situations.
- **Trustworthiness:** The 6G network will combine communication, sensing, processing, and intelligence, requiring a new network design. The new network design should support trustworthiness, collaborative sensing and distributed learning to enable AI applications. The 6G network architecture redesign is driven by data, knowledge, and intelligence produced from it, with new features that allow end-to-end trustworthiness. Novel data controlling frameworks and quantum attack detection technologies are included to improve privacy protection.
- **Sustainable development:** It is the ultimate objective of the 6G network design. 6G targets to increase the total energy efficiency 100 times throughout the network, preventing infrastructure and terminal energy consumption surpassing 5G while maintaining excellent service performance. 6G will significantly contribute to human sustainability (Saghezchi et al., 2022).

5.2. 6G future hardware and software perspective

While AI will play a significant role in 6G and Tactile Internet's expansion of services and capabilities, new concepts like self-

sustenance and evolution will be required in the network infrastructure. According to the 6G self-evolving network, the existing Hardware-Software (HW-SW) design techniques won't work for the research. The so-called HW-SW divide is a problem that has to be addressed. Increasing the isolation and symmetry between HW and SW are critical to enabling AI-driven 6G (Frei et al., 2013; Iannacci, 2021).

5.2.1. HW-SW divide

We've already established that the future 6G network must support a wide range of heterogeneous and exactly equivalent functions, including communications (ranging from gigahertz and below the THz range), computing, energy distribution, and various types of sensing, data collection, and storage for analysis and computation. AI-powered network will have to remain competitive with this. When the HW-SW gap is considered, such a situation displays its inherent limits, including the co-design method (Iannacci, 2021). The need to change the conventional bidirectional relationship between HW and SW is also growing. As stated in (Hautefeuille et al., 2009), Traditional joint design, which does not allow for rapid hardware modification, will not be used for 6G. We believe that a reformulation of this kind must include enhancements to our understanding of how HW and SW are separate but equally symmetrical.

5.2.2. HW-SW symmetry

An HW-SW separation strategy is required rather than the HW-SW co-design method. Both HW and SW are customized around a given set of functionality and specs. By addressing the heterogeneous, upgraded, and flexible HW capabilities required to integrate all of the mentioned services and support the autonomous system concept reinforced by AI. Increasing divergence between the HW and SW sectors will lead to a diversity of unique HWs and SWs changeable and unpredictable (Hafiz et al., 2016).

5.2.3. HW-SW separation

A rising HW-SW separation necessitates is essential to provide more excellent symmetry. Instead of being inflexible and incapable of self-adaptation and intelligence, the HW will need to develop these abilities to close the gap with SW. The 6G will seek a higher degree of complexity from the HW to relieve its present physical limits. It's important to note that AI is used broadly to include hardware's adaptability, reconfiguration, modifiability, and self-repairability.

5.2.4. MICRO-NANO technologies in 6G

The vast range of Micro-Nano technologies, such as electronics, MEMS/NEMS, and materials recognized to have significant potential for sparking the HW paradigm change outlined above, get a short perspective. Self-repairing and self-healing devices and materials may be realized using micro-and nanotechnologies. Carbon NanoTubes (CNTs) and microcapsule-filled healing agents are applied to repair electrically damaged electronics (Hafiz et al., 2016). Micro-fabrication methods enable the homogenous integration of sensing devices with various multi-functionality, enhancing the intelligence and adaptability of small pieces of HW. Logical circuits and highly integrated memory units may be realized using MEMS/NEMS latest technology, facilitating the distribution of computing capabilities to the edge. It is also important to note that micro/nano-fabrication enables the creation of effective passive radiofrequency. 6G and the IoT create demands that pose severe difficulties to HW-SW systems. Currently used methodologies to design and develop HW-SW systems pose a barrier to fully fulfilling the AI-driven network vision. The HW-SW gap was explored and potential solutions, including reformulating the relationships between HW and SW, are discussed (Necib et al., 2015). It is

claimed that Micro/Nano-technologies (MEMS/NEMS) would be a significant component in realizing changes in HW-SW system design (Uusitalo et al., 2021).

6. B5G and 6G vertical industries projects

Nowadays, all projects' main target is to lay long-term strategic roadmaps for the 6G wireless network. From 2017 to 2025, over €95 million has been assigned for B5G and 6G research investments, according to (Strinati et al., 2021). Most of these projects are funded by Horizon2020 and the EU's framework program for research and innovation. Also; most of them are still in their early stages. Fig. 10 highlights some of these projects. It also shows the investment in these projects concerning the total European fund. More than 156 organizations have participated. Each of them is attached to one project at least. Table 3 presents the main coordi-

nators, the number of participants, the total fund, and the project duration. Furthermore, more than half of the projects launched between 2020 and 2021, while more than 70% will end between 2022 and 2025, according to Fig. 10. The following subsections briefly illustrate some of the mentioned projects in Fig. 10 and Table 4.

6.1. B5G projects

The testing and validation of 5G technologies by integrators and vertical service providers is highly significant as 5G approaches maturity. Verticals have quite diverse demands. To take their services from ideas to prototypes and eventually products, they will need varying degrees of assistance. This section discusses the objectives, plans, and outputs of the most dominant beyond 5G projects.

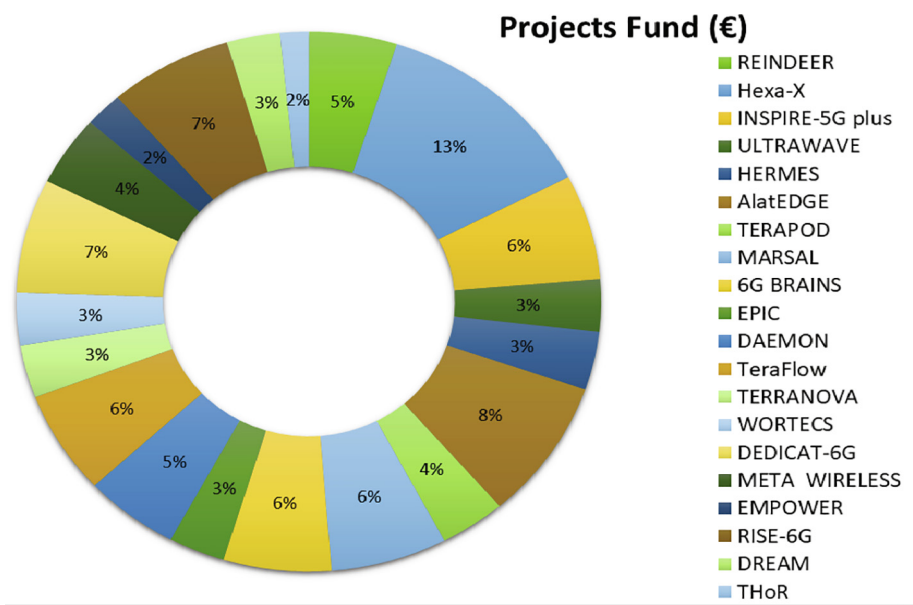


Fig. 10. 6G supported research projects and their fund information.

Table 4
Most important projects for B5G and 6G funded by Horizon 2020.

	Project name	Coordinator	Start date	End date	Fund (€)	Participants
B5G	MARSAL	Center Technologic De Telecomunicaciones De Catalunya - Spain	1 Jan. 2021	31 Dec. 2023	6,126,683.75	13
	TeraFlow	Center Technologic De Telecomunicaciones De Catalunya - Spain	1 Jan. 2021	30 June 2023	5,865,803.75	13
	DAEMON	Fundacion IMDEA Networks - Spain	1 Jan. 2021	31 Dec. 2023	4,997,025	11
	REINDEER	Technikon Forschungs und Planungsgesellschaft MBH - Austria	1 Jan 2021	30 June 2024	4,644,460	9
	THoR	Technische Universitaet Braunschweig - Germany	1 July 2018	31 Mar. 2022	1,498,877.5	NA
	EMPOWER	Sorbonne Universite - France	1 Nov. 2018	30 Apr. 2020	1,999,120	7
	TERRANOVA	University of Piraeus Research Center - Greece	1 July 2017	31 Mar. 2020	2,996,775	6
	EPIC	Technikon Forschungs und Planungsgesellschaft MBH - Austria	1 Sep. 2017	31 Aug. 2020	2,966,268.75	7
	WORTECS	Orange SA - France	1 Sep. 2017	31 Oct. 2020	2,999,155	6
	TERAPOD	Waterford Institute of Technology - Ireland	1 Sep. 2017	31 May 2021	3,467,593.75	NA
6G	ULTRAWAVE	University of Lancaster - UK	1 Sep. 2017	31 May 2021	2,971,366.25	8
	MetaWireless	Consorzio Nazionale Interuniversitario Perle Telecomunicazioni - Italy	1 Dec. 2020	30 Nov. 2024	3,995,128.44	11
	HEXA-X	Nokia Solutions and Networks - Finland	1 Jan 2021	30 June 2023	11,916,175	24
	DEDICAT-6G	Wings ICT Solutions Information & Communication Technologies IKE - Greece	1 Jan. 2021	31 Dec. 2023	6,499,613.75	13
	RISE-6G	Commissariat A L Energie Atomique ET AUX Energies Alternatives - France	1 Jan. 2021	31 Dec. 2023	6,499,613.75	12
	6G BRAINS	Institute for Research and Strategic Studies in Telecommunications GMBH - Germany	1 Jan. 2021	31 Dec. 2023	5,671,971.25	14

6.1.1. MARSAL project

For 5G and beyond (B5G), the MARSAL project will build and test a complete framework for network resource management using a converged optical wireless network architecture in the access/fronthaul/mid-haul segments. It proposes distributed processing cell-free massive MIMO designs connecting the current virtualized Radio Access Network (vRAN) parts and the Open RAN (O-RAN) concept to break down the limitations of the conventional cellular networks. For efficient and safe end-to-end data transmission, it offers a new paradigm for virtual infrastructures that integrates optical wireless convergence designs with networking, management, and security technologies in a transparent manner. For B5G networks, MARSAL's updated design provides new flexibility and closed-loop autonomy at all infrastructure tiers and dramatically improves spectrum efficiency through cell-free networking (<https://5g-ppp.eu/dedicat-6g/>).

A new generation of ultra-dense, cost-efficient, flexible, and secure networks may be enabled by MARSAL by focusing on three pillars: the network design pillar, virtual elastic infrastructure pillar, and network security pillar. Cell-free support will be added for the first time to O-RAN's architecture allowing for significant scaling up of Access Point (AP) deployment while also increasing performance. These network architectures will achieve high spectral efficiency and improved performance due to the significantly reduced inter-cell interference.

The elastic edge architecture supports dynamic slicing in the wireless and optical domains, which provides low perceived latency to Multi-access Edge Computing (MEC) applications. Using unique hierarchical network control planes and an elastic edge cloud architecture, MARSAL hopes to maximize network, compute and storage utilization and provide zero-perceived latency to innovative connectivity applications. Thanks to the network security pillar, MARSAL end-users and devices benefit from comprehensive network security and privacy approach in multi-functional environments. In multi-tenant scenarios, data security, integrity, and malicious traffic detection will be provided via innovative ML-based methods and a revolutionary blockchain-based platform that safely maximizes network flexibility.

6.1.2. TeraFlow project

An entirely new form of secure cloud-native SDN controller will be created by TeraFlow, which will revolutionize 5G networks. Using ML and Distributed Ledgers, this new SDN controller will be able to integrate the current NFV and MEC frameworks. This integration provides revolutionary features for both flow management (at the service layer) and optical/microwave network equipment (at the infrastructure layer).

Unifying network and cloud environments management, machine learning-based security, and decentralized consensus technological advances will be used to test the new TeraFlow architecture. The intended stakeholder pool now includes edge and hyper-scale cloud providers and conventional telecom carriers. By combining new and highly interactive network functions with zero-touch automation, these players will gain significant business agility. Communication and dissemination operations and strong engagement with standard defining organizations and the open-source software community will be significant parts of the TeraFlow project. A specific strategy for implementing TeraFlow findings will also be developed (Cherkaoui, 2021).

6.1.3. DAEMON project

DAEMON intends to achieve a systematic and pragmatic integration of Network Intelligence (NI) in mobile networks, guaranteeing that NI pushes zero-touch 6G systems towards achieving extremely high performance, sustainability, and reliability requirements. To achieve this aim, the network architecture must be

updated to support NI, and machine learning solutions must be created and deployed to fit more complex and diverse mobile networking environments.

On January 1, 2020, the DAEMON program was launched for the first time. The partners are presently working on the project's initial contributions (Vilalta et al., 2021).

Project's wide variety of network features and the methods to test their functional and non-functional needs. Furthermore, it is necessary to identify and address the present architectural models that prohibit a systematic NI integration and coordination in current standardization activities, such as ONAP, ETSI MEC, ETSI NVF, MANO, or O-RAN.

The DAEMON project's overall purpose is categorized into three technical goals:

1. Developing a B5G system architecture that is NI-native. While current attempts to integrate NI into mobile networks focus on adapting machine learning solutions to networking settings, DAEMON takes a different approach to modernize the network architecture to support NI operations natively. This objective is based on two pillars: (i) enhancing standardization of current architectural vision to enable comprehensive coordination across the numerous NI instances operating in the network infrastructure; and (ii) promoting a more profound and tighter integration of NI into the network infrastructure through the development and demonstration of novel coordinated NI solutions operating at the VNF level and implemented in the user plane.
2. Developing AI models that can trade off accuracy for network-critical metrics such as computational complexity and execution time in an adaptive manner.
3. Developing unique network features aided by NI for 6G systems. DAEMON specifies a list of critical network functions for which it will develop and implement NI algorithms capable of entirely using the proposed NI-native architecture and spanning several planes, domains, and operating timeframes.

6.1.4. REINDEER project

The REINDEER project will build a new intelligent connect-compute platform with scalable capacity and perceived zero latency while interfacing with many embedded devices. This is made feasible by the invention of Radio Weaves technology, a novel wireless access infrastructure comprised of a fabric of distributed radio, computer, and storage resources that acts as a vast distributed antenna array. Numerous future applications will significantly depend on wireless networking to enable real-time interaction with negligible latency and undetectable dislocation between virtual and physical parts. A remote medical professional might wear lightweight AR glasses with an integrated camera to monitor patient health data and manage medical equipment. The glasses are zero-energy devices that are wirelessly charged using signals supplied via the REINDEER platform. They receive high-definition video feeds for display purposes, but all processing is delegated to the infrastructure (Jiang and Schotten, 2021).

Similarly, humans and machines will collaborate in production, with a rising number of robots, autonomous vehicles, and drones. By 2030, integrated worlds in which interchangeable physical and virtual realities are possible. These applications will need little latency and a high degree of reliability. No retransmissions are permitted, and their working environment requires no downtime. Additionally, this will need accurate positioning technology at any place inside the coverage zone.

REINDEER's primary objective is analyzing and specifying technological needs for future interactive applications in industrial, entertainment, and health care use cases:

1. Adaptive robotic manufacturing, distribution, and logistics: Wireless connection will be crucial in future factories, warehouses, and logistics, with applications demanding zero-latency communication. Additionally, high-quality localization is required for robots, autonomous cars, and devices to cooperate. Cooperation between humans and robots in the exact location will need several concurrent broadband connections. Eventually, one would want to track all items using energy-neutral devices. Manufacturing and industrial environments are often highly reflective and include many obstructing items. As a result, future interactive apps in these contexts will confront complex constraints.
2. Crowd immersive entertainment: Crowds will want immersive entertainment enabled by augmented experiences and mixed-reality assistance. The future connection infrastructure must be scalable to handle a large volume of individual video services and better user experiences in densely populated areas, considerably beyond the present and developing wireless networks. This happens in huge, often open venues such as stadiums, festival grounds, vast halls, and auditoriums.
3. Natural human–machine interaction in care settings, hospitals, and assisted living facilities: Future human-oriented applications must enable intuitive experiences for humans engaging with robots and objects in their surroundings. Thus, the user-friendliness of the experiences is critical. The latter is true for both the application's dependability (humans may need assistance from robots) and the aesthetic integration of the electronics into the surroundings.

6.2. 6G projects

In this section, we choose the major 6G projects that launched recently, and we discuss the goals, objectives, and achievements of each project.

6.2.1. Hexa-X project

One of the major projects in 6G that was launched in 2021 is Hexa-X. Nokia launches Hexa-X with Ericsson (Ericson et al., 2021). Different research institutions and universities are cooperated to commercialize the latest technologies in this project. These include Dresden, Pisa, Aalto, Torino, Olu, Chalmers, Madrid, and Kaiserlautern. In addition, other commercial partners include Qacom, Networks, wings ICT Solutions, bcom, Atos, Intel, and Siemens, while TIM, Orange, and Telefonica are considered the operators in the project. By 2030, the Hexa-X project tends to put the foundation of the 6G networks to guide the research and Innovation (R&I) worldwide toward the next generation. The Hexa-X project seeks to develop tools necessary to bring the next generation of wireless communications to Europe and beyond. This project seeks to introduce new technologies and strategies for 6G to face six main challenges. We present the main 6G challenges in the Hexa-X project as follows:

- **Connecting intelligence:** Human-to-device communications must use emerging technologies like artificial intelligence (AI) and machine learning (ML) to increase the quality and efficiency of communications.
- **Network of networks:** The digital ecosystem requires establishing a unified network of networks. Ideally, this network must be versatile, intelligent, and scalable.
- **Sustainability:** For a sustainable network, resources should be exploited efficiently.
- **Global service coverage:** The 6G network's global and comprehensive coverage will require the development of cost-effective and practical approaches.

- **Extreme experience:** New network services need high data rate, low latency, huge capacity, and precision localization techniques.
- **Trustworthiness:** Next-generation systems should provide data privacy, the integrity of communications, confidentiality, and operational resilience for high security. Network design, artificial intelligence-driven air interface, THz radio access, and network virtualization will be developed to address these difficulties. The project will focus on bringing together the physical, digital, and human worlds via the use of these cutting-edge methods of communication. Hexa-X intends to create precise location, sensing, and resource management methods.

6.2.2. DEDICAT6 project

Dynamic coverage Extension and Distributed Intelligence for human Centric Applications with assured security, privacy, and Trust 6G project (DEDICAT6) is launched in 2021. In this project, several countries are evolved as Greece, Finland, France, Spain, the UK, and Germany. The main target of DEDICAT6 is to transmute B5G into an intelligent platform such as 6G. The vision of this smart platform needs to improve resources usage, reduce latency, decrease energy consumption, minimize capital and operational expenditures (CAPEX/OPEX), and strength security, trust, and privacy issues. The project will address four cases, including smart highway, smart warehousing, public safety, and enhanced experiences (Strinati et al., 2021).

DEDICAT6 project uses blockchain and AI techniques to lay an intelligent connectivity platform. This platform will allow the combination of the current infrastructure with an uncommon one. The new infrastructure should be intelligent, flexible, real-time supportive, and energy-efficient. The project introduces new techniques with predictive caching and intelligent computation placement for intelligent dynamic resource connectivity distribution. DEDICAT6 also seeks to develop and design mechanisms for dynamic coverage extension that will exploit connected devices such as vehicles, drones, and robots. These devices will cover the spot that cannot be accessed quickly in case of infrastructure, emergencies, or additional temporary capacity for a special event. Furthermore, many factors should be considered, such as de-vices locations and charging stations locations for optimally placing and distributing devices.

Moreover, security, trust assurance, and privacy are addressed for mobile services and the interface among digital ecosystems and human interaction. The project is expected to enhance significantly resource management, connectivity extension, security, intelligent network load balancing, and human–machine interactions. The project platform will set several solutions to achieve the project goals.

6.2.3. RISE 6G project

In January 2021, Reconfigurable Intelligent Sustainable Environments for 6G Wireless Networks was introduced as one of the primary technologies (RISE 6G). Seven countries and thirteen partners will adopt RISE-6G. We are proud to have CEA-Leti as one of our strategic partners. The project will continue working for the next three years (Di Renzo et al., 2019). The Reconfigurable Intelligent Surfaces (RIS) technology used in the RISE-6G is one of the most promising future. Controlling radio wave propagation in a dynamic and goal-oriented manner allows for the perception of wireless environments as services. An intelligent, sustainable, and adaptable wireless environment will be created using the RIS technology. A total of four difficulties will become the project's primary focus (Liaskos et al., 2018). Real-world signal propagation will be simulated first. Second, it will be integrated with different RISs

new network architecture. Third, numerous use cases will be developed to enable QoS in a dynamic wireless programmable environment, such as accurate localization, green communication, power consumption, and enormous capacity. Fourth, Specific rules governing radio airwaves and data security should be guaranteed.

Connected devices in high-capacity areas will be dynamically managed in RIS-enabled settings to minimize energy use to prevent interference. The project contributes to standardization and delivers its technical perspective to the execution of the project in the industrial sector. As a result, Europe will maintain its technological leadership in the B5G/6G global race, allowing for the development of new European-conceived services and economic opportunities.

6.2.4. Meta project

MetaWireless is another project funded by the EU. The project paves the way for the next generation 6G and beyond wireless networks. Commonly, most wireless networks systems assume that the nature of the propagation radio channel characteristics is static. Following generations of wireless networks consider that channels are uncontrollable factors (Chirivella-Perez et al., 2021). MetaWireless will treat the environment as a quantity to enhance and optimize the network performance. RIS will be provided to manipulate the wireless environment (Hussein et al., 2019). RIS are planar structures made of meta-materials and electromagnetically discontinuous. They don't involve the traditional laws of diffraction and reflection. However, they can be controlled to adjust the phase and wavefront of the radio waves. If deployed to coat objects, walls, or building facades, they could allow customizing the electromagnetic response of environments in real-time. The Integration of RISs and wireless networks will save more energy, enhance networks security, and increase degrees of freedom for system design. MetaWireless will develop the main foundations of algorithms, theoretical framework, optimization protocols, and network architecture applied RIS. Moreover, an open-access simulator for network optimization will be designed. The simulator will analyze ray tracing modules, build RIS environments, and test large-scale RIS environments (Ali et al., 2021).

6.2.5. 6G BRAINS project

Nowadays, future large-scale industrial tasks, services, assets, and devices need innovative wireless connectivity systems. These intelligent wireless connectivity systems need to be unlocked through new spectrum resources and AI systems to manage the network resources fully. Bringing Reinforcement learning Into Radio Light Network for Massive Connections (6G BRAINS) project (Shi et al., 2021) will evolve an AI-driven self-learning platform at multiple decision layers for future industrial applications of massive scale and varying demands such as the massive connections over device-to-device (D2D) (Hady et al., 2020). This AI system is designed for allocating resources dynamically in new spectrum links, including THz and Optical Wireless Communications (OWC). This system also aims to increase network reliability and capacity, enhance localization accuracy, and reduce end-to-end delay. The project is considered the first to propose a new cross-layer Deep Reinforcement-Learning (DRL) driven solution to support 6G requirements. The enabling technologies in 6G BRAINS discuss four main challenges as follows (Selem et al., 2019):

- Disruptive new spectral links enhancement such as THz and OWC.
- AI-driven D2D cell-free network modeling for vastly dynamic connections.
- Intelligent end-to-end network slicing while guaranteeing highly dynamic network QoS.

- AI with high precision 3D SLAM data fusion for indoor mapping with 1 mm positioning accuracy.

Several systems such as eHealth, intelligent transportation, Industry 4.0 will enable the technologies in 6G BRAINS (Selem et al., 2021). This project aims to set a fundamental standard in AI technologies for B5G and 6G. The project is expected to set a solid basis and global standardization for B5G and 6G technologies in fields related to industrial environments. The project solutions intend to convert traditional systems into intelligent, intelligent systems such as smart factories and intelligent transportation systems. These systems will have less static intelligent elements. For example, buildings will be considered fixed elements but intelligent and flexible. Each device in these novel environments can be wireless and routed with a 3D SLAM mechanism. The Multi-access Edge Computing (MEC) of 6G BRAINS will develop predictive mechanisms to be updated like the case of maintenance in old machines.

Moreover, modeling of D2D clusters' with AI systems is being developed to enhance resource sharing in the OWC band and the bands of 200 GHz and 300 GHz in 6G BRAINS. This modeling is vital to allow the usage of THz and OWC in 6G in the industrial environment. 6G BRAINS solutions (Nayak and Patgiri, 2021) also target reducing latency, enhancing communication service availability in case of high capacity, extending the coverage area for indoor, outdoor, and indoor/ outdoor environments, and enhancing concurrent transmission of non-real-time data and real-time streaming high critical data. Finally, this project will investigate devices mobility. It seeks to guarantee to extend coverage for all connected devices under several scenarios in all places such as factories or plants.

7. The 6G applications requirements and challenges

This section introduces the 6G promising applications and their challenges.

7.1. E-health

Nowadays, healthcare application is not dedicated only to patients; it has become mandatory to improve the modern life-style. Healthcare application includes emergency services, ambulance service, insurance service, Intelligent Wearable Devices (IWD), remote surgeries by doctors, Hospital-to-Home (H2H) service, pharmacies services, etc. One of these services is using IWD, which will help patients efficiently (Lu and Zheng, 2020). These devices are connected remotely and transmit required data about patients through the Internet. These data are collected and analyzed in monitoring and tests centers. The transmitted data by IWD is about heartbeat rate, health conditions, blood tests and pressure, the weight of the body, and amount of nutrition. IWD will keep patients' lives more manageable, reducing the number of hospitals visiting or staying at home for a long time. To enhance the performance of these devices, Wireless Body Area Networks (WBANs) are applied (Mucchi et al., 2020). WBANs collect the wearable devices' data in a heterogeneous environment. WBANs should prolong network lifetime and deliver data in emergency cases within accepted reliability to enhance performance (Saeidi et al., 2020).

Moreover, a vast capacity is necessary to connect the sensors and actuators in WBAN. Another service that is ultra-sensitive to a latency (<1 ms) is remote robotic surgeries (Khan et al., 2020). Remote robotic surgeries demand networks with extremely high reliability, data accuracy, high data rate to exchange information among two remote healthcare centers. To guarantee (QoS) require-

ments, healthcare networks 'shouldn't be defined as traditional wireless networks. This kind of network needs to have mobility support, tremendous capacity support, extremely low latency (<1 ms), green communication for patient safety, and continuous connection availability (99.99% reliability). 6G is considered the best solution to host healthcare networks for these requirements. 6G networks will revolutionize the healthcare sector. 6G is considered a solid candidate to support the health care network sector because of its reliability, mobility support, capacity support, and safety (Hussein and Abd El-Kader, 2017).

Moreover, 6G will be essential in offering an economical remote service for patients. Caring for a patient at home will be enabled by the 6G network. The network also will be able to deal with the massive number of patients. 6G is intended to be a supportive network that will turn the healthcare network true and evolve the human body as a part of the "Net." Healthcare network architecture in 6G is introduced in Fig. 11. This Fig shows the main deployment layers for this network, such as intelligent sensing, intelligent access, and intelligent cloud. In addition, Fig. 11 shows the major evolved technologies for healthcare services.

Using 6G networks in healthcare applications is discussed in (Bhat and Alqahtani, 2021). This paper aims to present an intensive overview of how 6G can improve the future of healthcare lifestyle based on several aspects such as technological aspect, sociological aspect, regulatory aspect, ethical aspect, etc. In addition, some challenges have been presented in the case of applying 6G networks for healthcare, such as security and resources allocation.

Paper (World Economic Forum, 2019) studied the importance of intelligent healthcare in our lifestyle. It means that healthcare will be fully AI-driven. The paper showed 6G as a promising communication technology for intelligent healthcare. Some of the supported applications in 6G have been investigated to meet this system soon, such as holographic communication, haptic technology, AI, AR, and VR. In addition, several new services have been suggested to enhance system quality as blood sample reader, H2H service, precision medicine, and a new system model for Healthcare Insurance at hospitals.

A meta-material-based wearable flexible Ultra-WideBand (UWB) antenna is designed (Lee et al., 2019). This antenna is designed for the WBAN-IoT application. It operates at sub-6G frequency bands. This antenna is considered to improve bandwidth efficiency, directivity, and gain. It is also designed to remove the stop bands by minimizing their size. To achieve these goals, the antenna is integrated with six suggested arrays that consist of modified grain rice shape within a Split Ring Resonator (SRR) and two capacitive-loaded strips (CLSs). Based on Performance analysis, the antenna performs good measurements compared to other antennas while showing smaller dimensions. The offered antenna presents BW expansion, stop-bands elimination, directivity gain enhancement.

Authors in (Katz et al., 2019) study the challenges of integrating IoT and WBANs in 6G. Many challenges face this kind of amalgamation, such as duplicated sensed data transmission, network lifetime, and routing efficiency in case of the limited number of available resources among sensor nodes in WBAN. Energy harvested and Cooperative-enabled efficient Routing Protocol (EHCPR) for IoT-WBAN in 6G is addressed to mitigate the mentioned challenges. The selection of the forwarder node for efficient routing is based on several parameters. These parameters are available bandwidth, signal-to-noise ratio (SNR) (Chen et al., 2019), sensor node's residual energy, number of hops to the destination, interference, and network congestion level. Compared to different WBAN protocols, EHCPR shows high performance due to the smart harvesting mechanism and dropping of redundant data packets from the transmission pool. The suggested protocol increases network reliability, latency, throughput, and lifetime.

7.2. Drones (unmanned vehicles)

An important aspect of wireless networks beyond 5G and 6G in the next decade will be the widespread and increasing usage of unmanned aerial vehicles (UAVs). Wireless communication must be dependable, secure, and cost-effective to enable such a vast deployment of UAVs. Cellular networks are critical to UAVs operat-

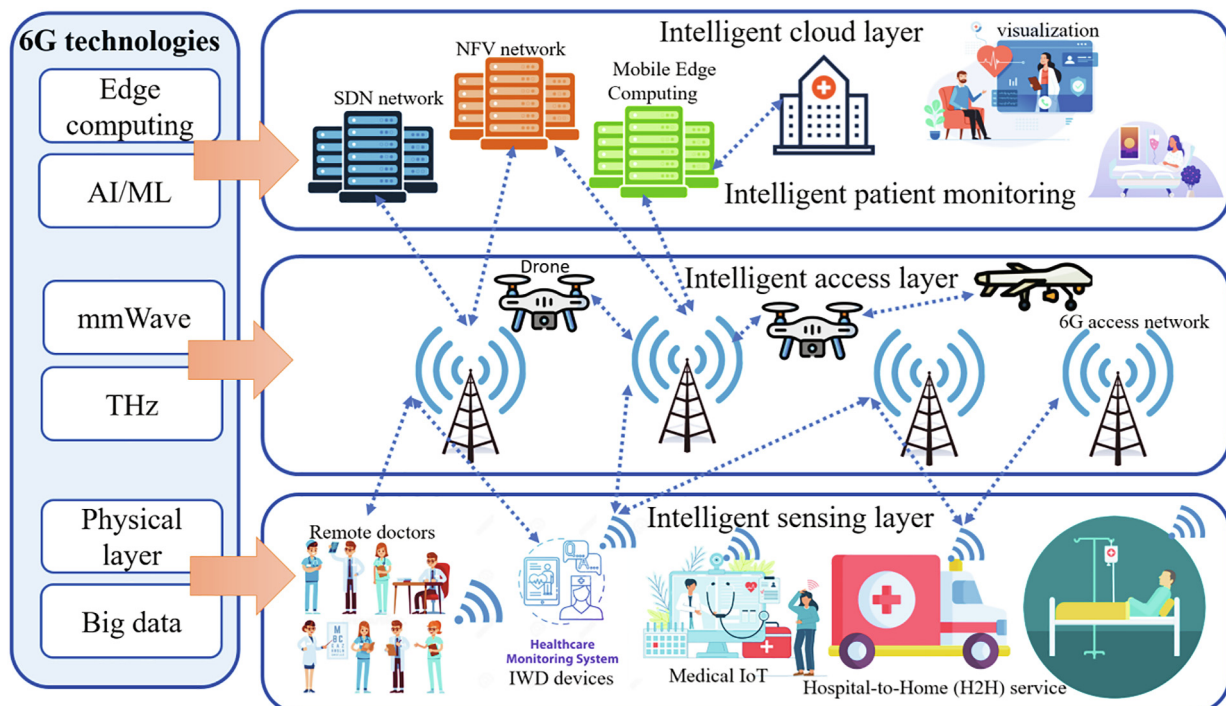


Fig. 11. Healthcare network architecture within 6G.

ing as flying user devices in this context. The cellular networks provide intriguing connection alternatives for UAVs; however, ensuring robust UAV operations confronts several difficulties. This paper shows an overview of the main design concerns and obstacles for broad commercial usage of flying UAVs and their possible solutions. We describe how cellular networks can enable UAVs using sophisticated features, network intelligence, essential enabling technologies beyond 5G and 6G, and machine learning techniques. Finally, we shed light on integrating non-terrestrial networks with terrestrial networks to achieve infinite connection in the 6G era of wireless services. UAVs may be used for package delivery, media creation, real-time surveillance, and distant building projects. The Federal Aviation Administration (FAA) states that commercial UAV fleets will reach 1.6 million by 2024.

Meanwhile, solutions for low-altitude UAVs' safe and secure operation with their significantly increased traffic density are being developed for UAS Traffic Management (UTM). In a linked sky, reliable wireless communications are necessary to facilitate the large-scale deployment of UAVs. This will need an adequate wireless communication infrastructure, clever calculation, and dependable control mechanisms. The following features may be included to assist UAV processes in different scenarios:

- 3D wireless connectivity with enhanced cellular networks and the incorporation of terrestrial and non-terrestrial networks,
- AI at the edge for UAV image processing in scenarios such as rescue, collision avoidance, and inspections, and robust control of and inter-UAV communication systems for flying in formation and cooperative operations are required.
- Wireless communication for UAVs flying at various altitudes may be provided through terrestrial cellular networks, perhaps with terrestrial and non-terrestrial networks.
- Cellular networks provide flying UAVs with wide-area, low-cost, and dependable wireless communication advantages.

On the other hand, the current cellular infrastructure is designed mainly to serve UEs situated on or near the ground. There

are significant issues in supporting aerial UEs because of their high altitude, mobility, large deployment, and flight safety regulations, especially in next-generation wireless networks that need to enable infinite connection. It has since garnered substantial interest in academia and industry in constructing cell-connected UAV systems for flying UEs (Zhang et al., 2020). Mobile-connected UAV research focuses on standardization, mobility, deployment optimization, and performance assessments (Hussein et al., 2020). Meanwhile, the 3GPP began standardization works for supporting aerial UEs in Release-15 and continued to work in succeeding 3GPP releases. There are various choices when using an unmanned aircraft system (UAS) to communicate with the outside world, UAV-UEs, for example, have more line-of-sight (LoS) channels than ground-based UEs because of their higher altitudes. LoS connections between UAVs and BSs are advantageous because they eliminate the signal obstruction and increase the received power of a reference signal. Interfering BSs may cause high LoS interference, which reduces the communication connection quality, even when such a LoS state is advantageous to the serving BS. Due to near-LoS channel conditions between BSs and UAVs, the cellular-connected UAVs suffer from serious interference in the downlink of their cellular connections. Similarly, uplink interference may be severe when there are many sending UAVs. Because of this, providing reliable communication in space is a challenge. UAV mobility and handover management in the skies are also tricky because of the comparatively fast speed of UAVs and the BS antenna layouts (Bi, 2019). In Fig. 12, UAV network architecture and use cases in 6G are addressed.

It is also possible to improve the performance of cellular connected-UAV networks by using the major enabling technologies of 5G and future generations.

- **Massive MIMO:** Ground BSs can conduct adaptive 3D beamforming with narrow beams to effectively serve UAVs in the sky and terrestrial users via the deployment of substantial antenna arrays of MIMO. To improve the spectral efficiency and coverage of UAV networks, massive MIMO may be used.

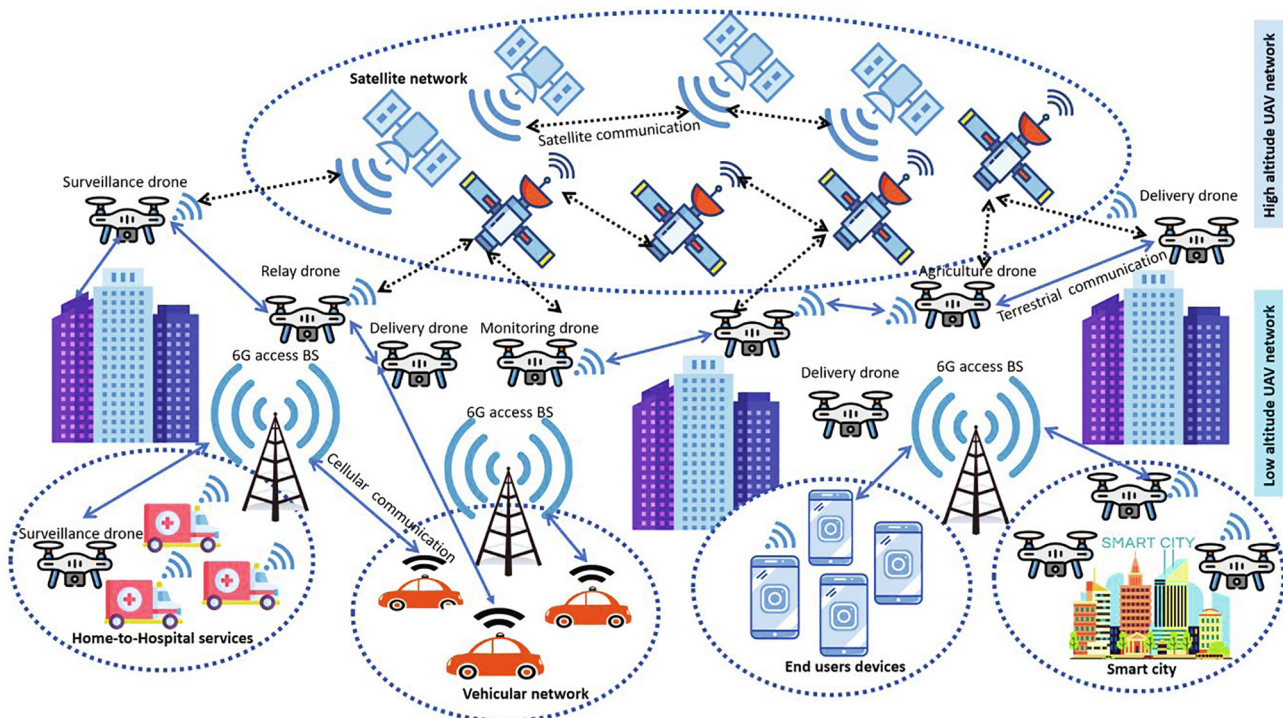


Fig. 12. UAV network architecture and use cases in future 6G communication.

Operation at high frequencies THz and millimeter-wave communications have the potential to provide high data rate services. In this paper (Bai et al., 2021), the authors presented a new stochastic geometry-based model for 6G large MIMO mmWave UAV channels. This model can characterize the influence of UAV-specific characteristics such as movement direction, height, and speed on channel statistical features. It also computes the space-frequency time-correlation function, envelope crossing rate, spectral density power, and fade length. The influence of UAV-related factors on channel statistical features is investigated, which may help designers of 6G massive MIMO mmWave UAV communication systems.

- **The high-frequency operation** provides a large amount of bandwidth. It enables massive MIMO systems to use small antennas, which are critical to satisfying the high data rate needs of UAV communications (Wang et al., 2020).
- **Intelligent Surfaces:** RISs automatically reflect electromagnetic signals in the desired direction to improve propagation conditions and communication connection quality. Cellular-connected UAV systems may also use RISs to handle the problem of down-tilted BS antennas by optimally reflecting sent signals toward UAVs in the sky and mitigating downlink interference, which can lead to increased coverage for UAVs. RISs have developed an energy-efficient and cost-effective technology to accommodate future services and needs. RISs generate a suitable propagation channel between the transmitter and receiver using many low-cost passives reflecting components. This paper (Basharat et al., 2021) introduces RISs for 6G, and it focuses on the performance benefits that may be gained by combining RISs with new communication technologies. The problems of implementing RIS-assisted networks, including RIS reconfiguration, deployment and size optimization, and channel estimates, are discussed in (Alghamdi et al., 2020). They also investigate RIS and NOMA with incomplete channel state information (CSI). Determining the size of RIS is a critical element in RIS-assisted networks. Finally, they outline possible future research avenues for 6G RIS-assisted networks.
- **Energy Harvesting:** Most devices applying 6G will dissipate much more energy than previous generations; because of the enormous capacity that will work in higher frequency bands. It is becoming more desirable to empower battery-limited devices by using energy harvesting and wireless energy transfer as technology advances towards 6G. Wireless energy harvesting will handle several scenarios for energy consumption: dedicated radio frequency harvesting sources, interference-aware harvesting, and ambient sunlight harvesting. Hence, intelligent energy-harvesting mechanisms are needed to eliminate energy consumption. UAVs with limited battery life may benefit from energy collection (such as solar and wind) and wireless power transmission through ground-based stations. In (Jiang et al., 2021), the UAVs are used for military and commercial uses due to their cheap cost, flexible deployment, and communication capabilities. UAVs may act as airborne base stations for terrestrial users, relays to link disconnected nodes or mobile users in cellular networks. However, tiny UAVs have limited onboard energy. UAVs can only be used to build temporary wireless communications. Low-power green UAV communication has become a significant challenge. This paper (Mao et al., 2021) does a complete study of 6G green UAV communications. Typical UAVs and their energy consumption models are described and normal green UAV communication trends. Also, the authors highlighted UAV uses and their green designs and mentioned potential methodologies and ongoing research questions.

7.3. Factory of the future

With the introduction of IoE and B5G/6G, machines, sensors, devices, robots, and even people will be connected inside factories. Additionally, human interaction will not exist for particular scenarios in some factories. Currently, most factories must be fast, intelligent, automated, and secure. This type of factory is called the “factory of the future” (Huang et al., 2019), as in Fig. 13. These factories can enhance the process of manufacture. They can balance between ‘market’s supplies and demand intelligently. They can predict ‘machines’ maintenance needs. Substantial global companies can monitor their factories remotely through global control centers. The companies also can predict the failure and success of a product line flow intelligently based on historically collected data (Huang et al., 2019). Covid-19 pandemic leads to rush the development of factory of the future concept to reduce the presence of the workers inside factories. Factory of the future is considered one of the challenging applications in the following wireless communication future. Advanced XR tools and human senses identification mechanisms are required to control factories remotely. Digital twins will be installed for machines, devices, robots, and networks supply. Massive data will be exchanged among devices and collected in control center operators. Several vehicles such as robots, drones, and cargo trucks need to be localized accurately.

Factories will be self-healing is a mandatory issue (Elzanaty et al., 2021). These challenges demand special wireless networks with high data rates, reliability, and low latency. Yet this requires solutions for new challenges, which at the level of the communication system are addressable by 6G. 6G will introduce the concept of D2D to overcome latency and capacity issues. 6G will adapt big data-based, real-time analytical applications techniques for digital twins (Xiao and Zeng, 2020). Furthermore, handling this dynamic and highly complex system will depend on advanced artificial intelligence mechanisms such as explainable AI, distributed AI, layered AI, etc. High-accuracy positing (<1cm) will rely on advanced localization techniques supported by the 6G network.

7.4. Localization

Due to increasing application requirements, traditional technologies of localization and sensing, like cell coordination and GPS, have become out-of-date (Bourdoux et al., 2020). Applications like self-driving cars and robotic surgery will need high-resolution positioning systems at the centimeter level in 2030’s next generation of wireless communication. Many mobile sensor-based apps that rely on location information should include location data, privacy and security policies, power consumption/unit distance coverage and routing strategies, alerts due to malfunctioning nodes, and other information about the area (De Lima et al., 2021). Other services, such as context-aware service apps, exploit the present location of a mobile user to provide more relevant services to that user. If you’re a driver, you may be interested in a real-time traffic report or perhaps a live video stream of your upcoming route. Real-world factors such as temperature, time, and location are examples of context. Automatic updates are needed for this information. Many scenarios where a specific network, such as 6G, should be used to meet these needs. The 6G network will automatically understand user needs and provide the optimum QoS (Wymeersch et al., 2021). Intelligence, ubiquitous networking, precise localization, and high-resolution sensing will be integrated into 6G systems to suit particular use cases around 2030, needing centimeter-level accuracy. Context-aware services will be able to connect, locate, and sense more efficiently due to this (Lv et al., 2020).

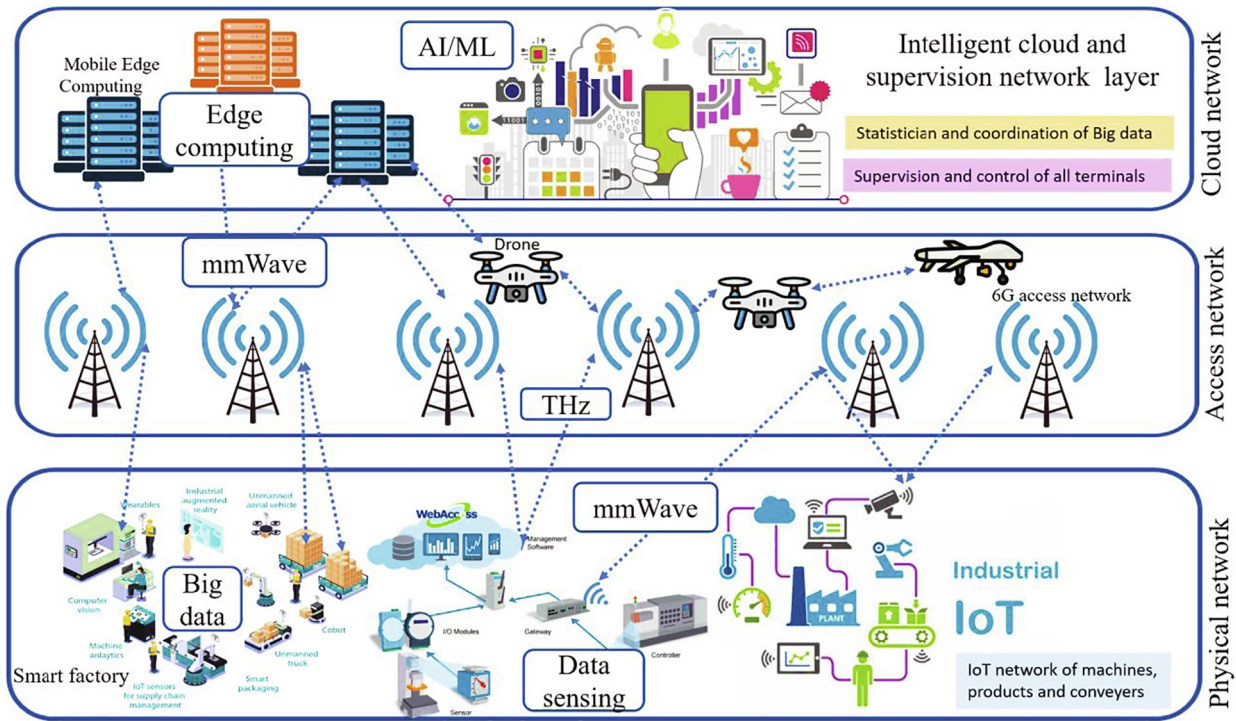


Fig. 13. Factory of the future technologies, requirements, and architecture within 6G.

Simultaneous Localization and Mapping (SLAM) mechanisms are addressed in (Hakeem et al., 2019). These mechanisms are divided into two tasks, localization, and mapping. They are based on self-perception by devices. This category of mechanisms is prevalent in robotics and ADAS. The paper presented a survey on Lidar SLAM and visual SLAM. The fundamental type of sensors and their stability, deep learning fusion, challenges, and open research issues are illustrated for Lidar SLAM or visual SLAM.

Furthermore, the integration between Lidar SLAM and visual SLAM has been highlighted. The integration addresses the multi-sensor calibration and hardware challenges. Due to the integration, SLAM mechanisms require a high data rate and less latency than 5G 'can't afford. Thus, this paper suggests 6G network deployment support such mechanisms.

Multi-Objective Function in 9D Localization: We can't simultaneously minimize the position, orientation, and speed estimation errors to achieve accurate 9D localization (Hakeem et al., 2020). Hence, the phase design turns into a multi-objective optimization problem, requiring more sophisticated algorithms. In this sense, efficient solutions for the RIS phase design by optimizing multi-objective functions are still lacking. For example, some use cases require that both the position and orientation errors be minimized to provide the user with more rich experiments such as AR applications. Furthermore, some communication-based metrics can be optimized (e.g., the coverage, rate, and energy efficiency).

Multiple waveforms may be generated for different time slots to estimate better different settings in a time-division approach for 9D localization. In addition, the 6G waveforms will be developed for effective interaction and localization, continuing a pattern initiated by LTE Advanced. The RIS phases and beamforming codebooks must be designed together at the BS and UE.

According to its intended purpose, a difference should also be established between RIS's employment as a transmitter or reflector. The difficulty of the joint optimization issue adds to the difficulty of the localization techniques. Reconstruction of the surroundings (holographic mapping) and self-location using the

rebuild map are possible with 6G's holographic simultaneous localization and mapping. Using holographic SLAM, users will better understand their surroundings and how they interact with them. The phase profile optimization issue may be solved using machine learning techniques. Machine learning algorithms employ a sub-sample of random sites to map the surroundings to map out the UE's position and orientation based on received signals. Despite this, the training data needed for data-driven methods using machine learning might be enormous to attain the desired accuracy. A machine learning strategy that doesn't demand many probable UE sites is preferable.

7.5. Connected autonomous vehicles (CAV)

Autonomous connected vehicles (CAVs) (Liu et al., 2020) are a critical vertical industry in 6G as it offers high-demand services to users. Autonomous vehicles (AVs) are vehicles that can operate with no human input. Among the most sophisticated forms of connected autonomous vehicles (CAVs) are those that can communicate and support infrastructure to enhance road safety and efficiency. The emergence of CAVs has drawn substantial momentum to address transportation issues. It can increase road safety, reliability, fuel consumption, and traffic congestion via several interesting CAV applications, including coordinated platooning. The more strict requirements for the connection of vehicles must be met by 6G to unlock the full capabilities of CAV. Extremely low latency (latency of 0.1 ms), massive connectivity, and ultra-high throughput to handle a large amount of data for complete automation are some features that make the system so reliable. New systems and communication technologies are needed to meet these requirements, such as C-V2X and 5G New Radio (NR) V2X.

CAV's essential services will benefit significantly from 6G. The primary use of 5G is for data transfer. 6G will enable new use cases more than just a telecommunications network. The CAV functions of communication, sensing, and positioning have been studied. 6G new THz technology with a shorter wavelength supports accurate

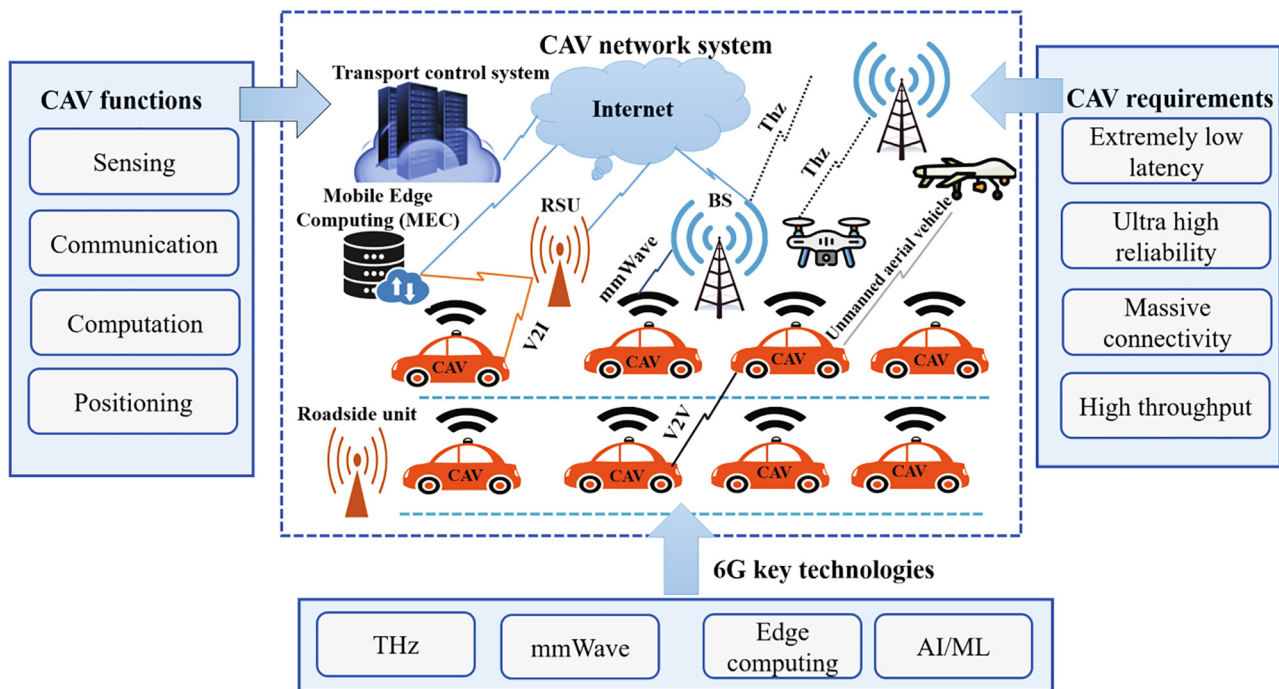


Fig. 14. Connected Autonomous Vehicles (CAV) functions, requirements, and architecture within 6G.

sensing and location. Increased data throughput may be achieved by using higher frequencies in beamforming. There is a growing trend toward using embedding intelligence inside communication networks to deal with the growing complexity of networks and network administration. 6G mobile systems can be implemented wide and cost-effectively to meet application needs. A more practical solution for all CAV tasks, which are also necessary for a system to perform self-driving functions, is expected via a combined development of a 6G system with various cross functions. The basic CAV network system consists of communication services, smart roads, roadside units (RSUs), and vehicles. To fulfill 6G's aim of wireless intelligence and decrease network operating costs, the unique CAV characteristics such as controlled mobility and simplicity of deployment may assist the CAV tasks by extending, monitoring, and maintaining 6G networks.

The CAV infrastructure receives enormous investments to build, manage, and operate services and resources in 6G-CAV communications. A CAV network function, requirements, and architecture within 6G are illustrated in Fig. 14. The CAV network consists of smart roads, roadside units, vehicles, and cellular infrastructure networks, all connected to the internet. The CAV networks offer some essential functions, including sensing of surrounding environment and road situations. Positioning and communication with other vehicles or infrastructure are key functions of CAV and computation tasks done in vehicles.

Autonomous driving is a primary focus for about 50 of the world's leading automotive manufacturers. Driverless vehicles will soon be available to the general public, and they will be reliable, safe, and eco-friendly. The introduction of CAV technology, such as driverless cabs and autonomous public transportation, creates a new service ecosystem. It is possible to categorize the many CAV security issues into three main categories: vehicle-level, supply chain-level, and data collection-level.

First, vehicle sensors, physical controls, and V2X communications are used to capture attacks on the vehicle level (Hakeem et al., 2021). Second, a lack of human involvement will lead to a physical takeover of the system. It is possible to include security measures in a vehicle. It is possible to send vehicle-triggered mes-

sages using 6G networks. CAVs may also be vulnerable to new types of V2X cyberattacks. Advance CAVs are linked to carmakers to continually check software changes and deliver to reduce any anticipated air issues (Hakeem and Kim, 2021).

However, the security and safety of automobiles and their passengers may still be compromised by communication channel flaws or cloud service providers misrepresenting data. Second, the CAV ecosystem has a complex supply chain that includes roadside equipment (RSE), cloud service providers, and regulators. Consequently, achieving uniformity in safety standards and interoperability is difficult.

CAVs might also face an issue of privacy when they record sensor data, travel itineraries, and the identities of their passengers. Protecting the privacy of 6G services is especially important while using public transportation modes like aircraft, trains, and buses. CAV security frameworks must thus consider security convergence by integrating physical and cyber safety (Hakeem et al., 2021; Hakeem and Kim, 2022).

8. Conclusion

Existing technologies are improved and new features are added when new generations of wireless communication networks are developed to satisfy future needs. Thus, 5G technology cannot handle the ever-increasing communication demands. Therefore, it is necessary to plan for the deployment of 6G networks to meet the needs of this new age in terms of communication technology. This paper presents an intensive overview for the 6G wireless generation networks including the growth of communication networks from 1G to 5G, research efforts on 6G mobile networks, enabling technologies for 6G network, and current state-of-the-art. The paper also highlights the primary necessities for 6G enhancement as high coverage, network flexibility, cognitive network, network computing, and trusted systems.

Leading projects for B5G and 6G, funded by European Union, are introduced briefly. The newest application and services like health-care networks, drones, autonomous vehicles, and smart factories

are discussed in this study. Also, the 6G hardware-software evolution perspectives are discussed compared with the conventional Hw-SW codesign. The paper also introduces the 5G timeline evolution and 5G advanced features. We also discussed the 6G promising applications and their expected structures and evolved technologies. We surveyed the most recent 6G related works and provided a comprehensive survey of 6G standardization, deployment, scenarios, and industrial projects. We wish that this study will attract more research into the enabling technologies for 6G networks and IoT applications in the face of a widening digital gap.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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